



TEACHING COMPANION · VERSION 3 · CHAPTERS 1–40

YMnotes Digestive System

Integrated Teaching Extras

Everything the three extras handouts add to the main book, merged into one book and filed under the chapter it belongs to. Teach a chapter from the book, then turn to the same chapter here.

Part 1 · Before mid-module

Chapters 1–15 · book file 1, pp. 1–197
Chapter 15 continues to p. 209 in file 2

Part 2 · After mid-module

Chapters 16–40 · book file 2, pp. 197–433

INTERNAL MEDICINE

PHARMACOLOGY

PATHOLOGY

MICROBIOLOGY

PARASITOLOGY

HEPATOLOGY

149 teaching additions · 36 points for review

From GIT Internal Med, GIT Others and the Hepatology source
Keyed to the printed page numbers of the main book



YMNOTES



INTERACTIVE CONTENTS · ALL 40 CHAPTERS

Chapter by chapter

How to use it. Teach a chapter from the main book, then open the same chapter here: every point adds something that is **not** on those book pages. Each point ends with its ID, **Book p.** (the printed page it extends) and its source page. Click a chapter to jump to it; **↑ Contents** and the running header bring you back. Source disagreements are not taught as facts; they are in **Points for Youssef to Review**.

SUBJECT LABELS

INTERNAL MEDICINE

PHARMACOLOGY

PATHOLOGY

MICROBIOLOGY

PARASITOLOGY

HEPATOLOGY

PART 1 · BEFORE MID-MODULE Chapters 1–15 · book file 1 (printed pp. 1–197); Chapter 15 runs on to p. 209 in file 2

CH	BOOK CHAPTER · SUBJECTS	BOOK PAGES	IN THIS HANDOUT	PAGE
01	GI Symptoms and Investigations Internal Medicine · Radiology / Investigation	Book pp. 1–9	9 additions	3
02	The Esophagus Internal Medicine · Surgery	Book pp. 10–32	1 addition	5
03	The Stomach and Duodenum Internal Medicine · Pharmacology · Surgery	Book pp. 33–69	18 additions · 6 review	6
04	Pathology of GIT Tumours Pathology · Surgery	Book pp. 70–75	8 additions	10
05	Diarrhea and Malabsorption Internal Medicine	Book pp. 76–94	15 additions · 2 review	12
06	Constipation Internal Medicine	Book pp. 95–105	4 additions · 3 review	17
07	Foodborn Infection Microbiology · Internal Medicine	Book pp. 106–114	6 additions · 1 review	18
08	Inflammatory Bowel Disease Internal Medicine · Pharmacology	Book pp. 115–121	6 additions · 2 review	19
09	Parasitic GI Diseases Parasitology	Book pp. 122–139	No additions	21
10	Small Intestine and Colon Surgery Surgery	Book pp. 140–144	No additions	21
11	Diverticulosis of the Colon Surgery · Internal Medicine	Book pp. 145–150	No additions	21
12	Appendix Surgery	Book pp. 151–161	No additions	21
13	Intestinal Obstruction Surgery · Emergency / Acute Care	Book pp. 162–188	No additions	21
14	Polyposis of Colon Surgery · Pathology	Book pp. 189–194	1 addition	21
15	Colorectal Cancer Surgery · Pathology	Book pp. 195–209	3 additions · 1 review	22

PART 2 · AFTER MID-MODULE Chapters 16–40 · book file 2 (printed pp. 197–433)

CH	BOOK CHAPTER · SUBJECTS	BOOK PAGES	IN THIS HANDOUT	PAGE
16	Anorectal Diseases Surgery	Book pp. 210–230	No additions	23
17	Hepatology and Hepatic Surgery Surgery	Book pp. 231–235	2 additions	23
18	Liver Disease Assessment Radiology / Investigation · Internal Medicine	Book pp. 236–250	18 additions · 3 review	23



19	Acute Hepatitis Internal Medicine	Book pp. 251–263	7 additions · 1 review	27
20	Chronic Hepatitis B Internal Medicine · Pharmacology	Book pp. 264–273	9 additions · 5 review	28
21	Autoimmune Hepatitis Internal Medicine	Book pp. 274–279	2 additions · 2 review	30
22	Liver Cirrhosis Internal Medicine · Pathology	Book pp. 280–307	8 additions · 5 review	30
23	Portal Hypertension Internal Medicine · Surgery	Book pp. 308–317	7 additions · 3 review	32
24	Hydatid Cyst Parasitology · Surgery	Book pp. 318–323	No additions	33
25	Pyogenic Liver Abscess Surgery · Internal Medicine · Parasitology	Book pp. 324–331	No additions · 1 review	33
26	Hepatocellular Carcinoma Surgery · Pathology	Book pp. 332–336	No additions	33
27	Ascites Internal Medicine	Book pp. 337–344	12 additions	34
28	Renal Impairment in Cirrhosis Internal Medicine	Book pp. 345–348	4 additions · 1 review	36
29	Jaundice Internal Medicine	Book pp. 349–354	4 additions	37
30	Hepatic Encephalopathy Internal Medicine	Book pp. 355–360	4 additions	38
31	Acute Hepatic Failure Internal Medicine · Emergency / Acute Care	Book pp. 361–366	1 addition	38
32	Biliary System Anatomy Surgery	Book pp. 367–372	No additions	39
33	Gallbladder Diseases Surgery	Book pp. 373–375	No additions	39
34	Gallstones and Cholecystitis Surgery	Book pp. 376–388	No additions	39
35	Obstructive Jaundice Surgery · Radiology / Investigation	Book pp. 389–398	No additions	39
36	Acute Pancreatitis Surgery · Emergency / Acute Care	Book pp. 399–406	No additions	39
37	Cancer Pancreas Surgery · Pathology	Book pp. 407–412	No additions	39
38	Acute Abdomen Emergency / Acute Care · Surgery	Book pp. 413–417	No additions	39
39	Abdominal Trauma Surgery · Emergency / Acute Care	Book pp. 418–424	No additions	39
40	Splenic Trauma Surgery · Emergency / Acute Care	Book pp. 425–433	No additions	39

AFTER THE CHAPTERS Kept apart from the teaching points

R	Points for Youssef to Review 36 source disagreements · both positions, with page references · not resolved	40
M	Sources, method and traceability The three handouts, page conventions, what was merged or left out	46



PART 1 · BEFORE MID-MODULE

Chapters 1–15

Book file 1, printed pp. 1–197. Chapter 15 (Colorectal Cancer) starts on p. 195 and runs on into book file 2 to p. 209; printed p. 197 appears in both files. It is treated here as one chapter.

CHAPTER 01 BEFORE MID-MODULE

[↑ CONTENTS](#)

GI Symptoms and Investigations

Adds to book pp. 1–9 · book file 1 · book subjects: Internal Medicine · Radiology / Investigation

IN THIS CHAPTER Symptoms and signs · Investigations

Symptoms and signs BOOK PP. 2–5

Why alarm features matter

INTERNAL MEDICINE

Many GI consultations are for functional (non-organic) symptoms, yet about **20% of all cancers** arise in the GI tract. The clinician's main task is therefore to separate organic disease from functional symptoms, and alarm features are the tool for doing so.

E01.01 · Book p. 2 · GIT Internal Med p. 1

Vomiting: how it is triggered

INTERNAL MEDICINE

- ◆ **Vomiting centres** lie in the reticular formation of the medulla. They are stimulated by the **chemoreceptor trigger zone (CTZ)** and by the **vagus**; drugs, motion sickness and metabolic causes can stimulate these zones directly.
- ◆ Vomiting is usually preceded by two phases:
 - **Nausea**, with autonomic features (salivation, pallor, sweating).
 - **Retching**, a strong, involuntary, unproductive effort to vomit with contraction of the abdominal muscles.
- ◆ Add **vestibular disease** to the CNS causes of vomiting (alongside raised intracranial pressure and migraine).

E01.02 · Book p. 3 · GIT Internal Med p. 2

Flatulence

INTERNAL MEDICINE

Flatulence means excessive wind: belching, abdominal distension, gurgling, or passing flatus per rectum. Increased gas production comes from colonic bacterial breakdown of non-absorbed carbohydrate, a high-fibre diet, and carbonated drinks.

E01.03 · Book p. 3 · GIT Internal Med p. 2



Abdominal pain: extra diagnostic clues

INTERNAL MEDICINE

- ◆ **Porphyria** is another metabolic cause of an apparent "acute abdomen", alongside diabetic ketoacidosis.
- ◆ **Epigastric pain:** functional dyspepsia is the commonest diagnosis, but its symptoms can be **identical** to those of peptic ulcer disease.
- ◆ **Biliary pain** can be felt in the epigastrium as well as the right hypochondrium. It remits spontaneously and recurs weeks or months later.
- ◆ **Peptic ulcer** sometimes presents with right hypochondrial pain.
- ◆ Chronic, constant hypochondrial pain (right or left) in a **well-looking patient** is frequently functional. Right iliac fossa pain may also be functional.

E01.04 · Book p. 4 · GIT Internal Med p. 4

Blood per rectum: where is it from?

INTERNAL MEDICINE

- ◆ Dark blood and clots per rectum **without shock** always come from a lower GI source. An upper GI bleed produces blood per rectum only when it is massive, and then it is usually accompanied by shock.
- ◆ Haematochezia usually comes from a lesion distal to the **ligament of Treitz** (duodenojejunal junction). It can also occur with a massive upper GI bleed because of rapid transit.

E01.05 · Book p. 4 · GIT Internal Med pp. 5, 37

Investigations BOOK PP. 5–9

Endoscopy: practical details

INTERNAL MEDICINE

- ◆ **Sigmoidoscopy** is also used for diarrhoea and change in bowel habit. The **rigid** scope inspects at most 20–25 cm of distal colon; the **flexible** scope can reach the splenic flexure.
- ◆ **Colonoscopy:** benign strictures are **dilated**; malignant strictures are **stented**.
- ◆ **Balloon enteroscopy** (single or double balloon) can examine the small bowel from the duodenum to the ileum, using specialised enteroscopes in expert centres.
- ◆ **Capsule endoscopy:** avoid it if a stricture is suspected.

E01.06 · Book pp. 6–7 · GIT Internal Med pp. 6–7

Imaging: extra interpretation points

INTERNAL MEDICINE

- ◆ **Plain films:** chest as well as abdominal films are used in the acute abdomen; interpretation depends on gas shadows inside and outside the bowel. Plain films can also assess **faecal loading** in constipation. CT is more sensitive than plain films for gallstone and chronic pancreatitis calcification.
- ◆ **Ultrasound** in the acute abdomen can also show enlarged mesenteric glands.
- ◆ **CT:** oral intraluminal contrast (e.g. Gastrografin) with IV contrast improves diagnosis. CT is sensitive to **small volumes of gas** from a perforated viscus and to **contrast leak** from the gut lumen. In **high-grade** obstruction it is usually diagnostic of both the presence and the cause. It can guide biopsy of tumours or lymph nodes.
- ◆ **PET** is also used to detect **recurrent** disease.

E01.07 · Book pp. 7–9 · GIT Internal Med pp. 7–8

**CT colonography: how it is done**

INTERNAL MEDICINE

CT colonography ("virtual colonoscopy") is performed after CO₂ insufflation into a previously cleansed colon. It is an alternative to colonoscopy for diagnosing colonic mass lesions.

E01.08 · Book p. 9 · GIT Internal Med p. 8

Contrast studies

INTERNAL MEDICINE

- ◆ Gastroscopy replaced the barium meal because it is **more sensitive** and allows **biopsy**.
- ◆ **Small bowel follow-through** shows the fold pattern and calibre of the small bowel; dedicated views of the terminal ileum show **early changes of Crohn's disease or TB**.
- ◆ **Small bowel enema (enteroclysis)**: a tube is passed into the duodenum and a large volume of dilute barium is introduced, to show strictures or adhesions when intermittent obstruction is suspected. It has generally been replaced by **MR enteroclysis**.

E01.09 · Book p. 9 · GIT Internal Med p. 9

CHAPTER 02 BEFORE MID-MODULE

↑ CONTENTS

The Esophagus

Adds to book pp. 10–32 · book file 1 · book subjects: Internal Medicine · Surgery

Hiatus hernia: surgery BOOK P. 20**Two further indications for surgery**

INTERNAL MEDICINE

Besides failure of medical treatment, two further indications for anti-reflux / hiatus hernia repair:

- ◆ a **large** hiatus hernia;
- ◆ **complications**: ulceration, severe inflammation, bleeding, stricture or Barrett's oesophagus.

E02.01 · Book p. 20 · GIT Internal Med p. 17



The Stomach and Duodenum

Adds to book pp. 33–69 · book file 1 · book subjects: Internal Medicine · Pharmacology · Surgery

IN THIS CHAPTER Anatomy, physiology and gastritis · Helicobacter pylori · NSAIDs, erosive gastritis and the drugs · Acute upper GI bleeding · Acute lower and chronic GI bleeding

Anatomy, physiology and gastritis BOOK PP. 34–38, 42

Gastric mucus and the functions of acid

INTERNAL MEDICINE

- ◆ Mucus-secreting cells are present **throughout the stomach** and secrete both mucus and bicarbonate. The mucus is made of glycoproteins called **mucins**.
- ◆ Gastric acid is **not essential for digestion**, but it protects against some food-borne infections.
- ◆ Other gastric functions: a **reservoir** for food, **emulsification of fat** and mixing of gastric contents.

E03.01 · Book p. 35 · GIT Internal Med pp. 21–22

Classifying gastritis and its causes

INTERNAL MEDICINE

- ◆ Gastritis may be classified by **acuity** (acute vs chronic), by **histology**, or by **etiology**.
- ◆ **Gastropathy** (epithelial injury and regeneration with little inflammation) is usually caused by **bile reflux**, **hypovolaemia**, or **chronic congestion** as in portal hypertension.
- ◆ Named causes of gastritis:
 - H. pylori gastritis, the **commonest** cause
 - erosive and haemorrhagic gastritis (NSAIDs)
 - autoimmune gastritis
 - viral infection (CMV, herpes simplex)
 - Crohn's disease

E03.02 · Book pp. 35–36 · GIT Internal Med p. 23

Acute gastritis: extra points

INTERNAL MEDICINE

- ◆ Stress gastritis also occurs in **acute myocardial infarction** and **stroke**, as well as after surgery, trauma, burns and CNS injury.
- ◆ Most acute gastritis settles within days to weeks, but some cases become recurrent or long-term, i.e. **chronic gastritis**. Over time this raises the risk of gastric ulcer, upper GI bleeding and some gastric neoplasms.

E03.03 · Book p. 36 · GIT Internal Med pp. 24–25

Where intestinal metaplasia comes from

INTERNAL MEDICINE

Gastric metaplasia, usually of the **intestinal type**, occurs almost always in the setting of **atrophic gastritis**. The book's Chapter 4 (p. 72) says intestinal-type gastric cancer arises on intestinal metaplasia; this supplies the missing first link. Chapter 4 here completes the sequence to dysplasia and cancer.

E03.04 · Book pp. 38, 42 · GIT Internal Med p. 26



Helicobacter pylori BOOK PP. 40–41, 48

Why combination therapy, and who is treated

INTERNAL MEDICINE

- ◆ **No single drug** is effective against *H. pylori* on its own, so regimens combine two antibiotics with powerful acid suppression.
- ◆ Resistance to metronidazole and clarithromycin is high in some populations, and oral metronidazole frequently causes side-effects.
- ◆ **All** patients with duodenal or gastric ulcer who have *H. pylori* should receive eradication therapy. Treating incidental infection with **no ulcer** is controversial.

E03.05 · Book pp. 40–41, 48 · GIT Internal Med p. 31

NSAIDs, erosive gastritis and the drugs BOOK PP. 36–37, 44–48

NSAID damage: how common, and why it can be missed

INTERNAL MEDICINE

- ◆ Aspirin inhibits the synthesis of both prostaglandin E2 and **F2-alpha**.
- ◆ With recurrent NSAID use (e.g. osteoarthritis) erosive gastritis can become **chronic**, and in alcoholics and chronic NSAID users it may be **completely asymptomatic**.
- ◆ Only about **5%** of NSAID users have GI symptoms and **1–2%** a major problem such as a GI bleed.

E03.06 · Book pp. 36–37 · GIT Internal Med pp. 32, 35

Drug options for acute and erosive gastritis

INTERNAL MEDICINE

- ◆ Three drug options for erosive and haemorrhagic gastritis: **H2 blockers** (e.g. famotidine, which may be used along with the PPI), **proton pump inhibitors** and **sucralfate**. In the book, sucralfate appears only in the pharmacology table (mainly stress-ulcer prophylaxis).
- ◆ Remove risk factors: stop NSAIDs, reduce caffeine, stop alcohol and smoking.

E03.07 · Book pp. 36, 48 · GIT Internal Med pp. 25, 32

Long-term PPI use and bone

PHARMACOLOGY

Osteoporosis is a possible complication of chronic (long-term) PPI use. Add it to the PPI adverse effects, after vitamin B12 deficiency.

E03.08 · Book p. 45 · GIT Others p. 11 (MCQ 24, keyed on p. 12)

Acute upper GI bleeding BOOK PP. 50–52

Drugs and bleeding: two qualifiers

INTERNAL MEDICINE

- ◆ **Even 75 mg/day of aspirin** can produce ulcers, erosions and GI haemorrhage.
- ◆ GIT Internal Med states that corticosteroids in the usual therapeutic doses have **no influence on GI haemorrhage** on their own. (Book p. 50 still advises avoiding NSAID–corticosteroid combinations.)

E03.09 · Book p. 50 · GIT Internal Med p. 37



Natural history and prognosis

INTERNAL MEDICINE

- ◆ Any significant GI bleed within the last **48 hours** should be assessed in hospital.
- ◆ About **85%** of patients stop bleeding spontaneously within 48 hours; many need no immediate treatment because the blood loss is small.
- ◆ Mortality from GI haemorrhage has stayed at **5–12%** despite advances in management, mainly because patients are now older and have more co-morbidity.

E03.10 · Book p. 50 · GIT Internal Med pp. 37–38, 48

Resuscitation and history: practical triggers

INTERNAL MEDICINE

- ◆ **Circulation:** unless there are clear signs of a cardiac cause, give IV fluid to any patient with **cool peripheries and a fast heart rate**.
- ◆ **Disability:** common causes of unconsciousness are profound hypoxia, hypercapnia, cerebral hypoperfusion, or recently given sedatives/analgesics.
- ◆ **History:** co-morbidity; acid-peptic symptoms; alcoholic liver disease, chronic hepatitis or cirrhosis; anticoagulants, antiplatelets, NSAIDs, steroids, alcohol binge; coagulation disorders. Ask about **epistaxis or haemoptysis** to exclude a non-GI source of the blood.

E03.11 · Book p. 50 · GIT Internal Med p. 38

Transfusion and supportive care

INTERNAL MEDICINE

- ◆ Monitor pulse and BP **half-hourly**; cross-match **2 units** initially. Give plasma expanders or 0.9% saline until blood is available, then restore volume rapidly, preferably with red cell concentrates through large-bore cannulae.
- ◆ **Shock** is a trigger for transfusion: pallor, cold nose, systolic BP < 100 mmHg, pulse > 100/min. (The source's haemoglobin threshold is kept for review in R05, not taught here.)
- ◆ **Pulse rate and venous pressure** are the best guides to adequate transfusion (the book, p. 51, explains why Hb is not). Insert a **CVP line** if there is organ failure or a high risk of heart failure, and watch for fluid overload, especially in the elderly.
- ◆ Manage patients in **high-dependency beds**.
- ◆ GIT Internal Med uses a **nasogastric tube** for gastric wash and monitoring of ongoing bleeding in non-variceal bleeding. The book is silent here; for variceal bleeding it treats the tube as selective (p. 315, review point R33).

E03.12 · Book pp. 50–51 · GIT Internal Med pp. 39–40

Stigmata of recent haemorrhage

INTERNAL MEDICINE

- ◆ Endoscopy finds the cause in **80% or more** of cases.
- ◆ Stigmata of recent bleeding are a **spurting vessel, active oozing, fresh or organised clot, or black spots**. Their presence means a **higher risk of rebleeding**, which is why they are an indication for endoscopic haemostasis.

E03.13 · Book p. 51 · GIT Internal Med p. 40



Endoscopic haemostasis: how each method works

INTERNAL MEDICINE

- ◆ **Why combine:** dual or triple therapy reduces rebleeding more than monotherapy.
- ◆ **Adrenaline injection** gives temporary local tamponade and vasoconstriction: 0.5–1.0 mL aliquots of epinephrine 1:10,000 into four quadrants of the ulcer, within 1–2 mm of the bleeding point. Used **alone** it is less effective than other monotherapies. Its main use in active bleeding is **pre-injection**, to slow bleeding and improve the view before definitive therapy. Sclerosants (e.g. ethanolamine) and thrombogenic agents are less efficacious.
- ◆ **Thermal:** contact probes compress the vessel and then coagulate it (coaptive coagulation); the **bipolar probe** is the most widely used. **Argon plasma coagulation** is non-contact: monopolar current delivered through ionised argon gas. Other options are the heater probe and laser.
- ◆ **Clips** are the therapy of choice for **obvious arterial bleeding**. Unlike cautery they cause no further tissue damage, and they reduce rebleeding.

E03.14 · Book p. 51 · GIT Internal Med pp. 40–42

Specific causes and after-care

INTERNAL MEDICINE

- ◆ **Mallory–Weiss tear:** caused by a sudden rise in intra-abdominal pressure. A history of retching **may be absent**. Most bleeds are minor and patients are usually discharged within 24 hours, though the haemorrhage can occasionally be large.
- ◆ **After a bleeding ulcer:** continue the PPI for **4 weeks**; always **confirm H. pylori eradication**; if eradication fails, give long-term acid suppression.
- ◆ **Bleeding gastric cancer:** bleeds are usually not large, but oozing from the tumour is very difficult to control endoscopically. Surgery is occasionally needed for uncontrolled or repeated bleeding and can usually wait until full evaluation. Radiotherapy is occasionally successful.

E03.15 · Book pp. 50–52 · GIT Internal Med p. 45

Antithrombotic drugs during a GI bleed

INTERNAL MEDICINE

- ◆ Clopidogrel and warfarin are also stopped, but stopping antiplatelets needs urgent discussion with a **cardiologist** (the book, p. 50, gives the thrombosis risk).
- ◆ **After PCI**, heparins and clopidogrel increase GI bleeding: give an **IV PPI bolus followed by an infusion**; **platelet infusion** can counter clopidogrel.

E03.16 · Book p. 50 · GIT Internal Med pp. 40, 45

Acute lower and chronic GI bleeding BOOK PP. 52–53

Lower GI bleeding: sorting the source

INTERNAL MEDICINE

- ◆ **Unstable patient:** GIT Internal Med performs **nasogastric lavage** (see R33 on nasogastric tubes). Blood in the aspirate identifies an upper GI source; a clear aspirate makes a lower GI source more likely. Then proceed to angiography with embolisation, endoscopic intervention or, in extreme cases, surgery. Each of these both locates and treats the bleeding.
- ◆ **Stable patient:** abdominal and pelvic CT with oral contrast is used to look for the cause.
- ◆ **Angiography** has a low yield, so it is a test of last resort.

E03.17 · Book pp. 52–53 · GIT Internal Med pp. 46–47



Chronic GI blood loss

INTERNAL MEDICINE

- ◆ The cancers to exclude are particularly **stomach** and **right colon**.
- ◆ **Occult blood tests are unhelpful** in this work-up.
- ◆ Oesophageal varices usually bleed overtly and **rarely** present as chronic blood loss. **Hookworm** is the most common cause worldwide.
- ◆ Do upper endoscopy and colonoscopy **in the same session**. A colonic polyp found at colonoscopy should **not** be assumed to be the cause.
- ◆ In frail patients, **unprepared CT** is a reasonable test for colon cancer. Capsule endoscopy yields a diagnosis in **60–85%** but cannot treat.

E03.18 · Book p. 53 · GIT Internal Med p. 48

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R01 Autoimmune gastritis: is it "pangastritis"?** (p. 40) · **R02 Misoprostol: a PGE₁ or a PGE₂ analogue?** (p. 40) · **R03 H. pylori triple therapy: duration and first-line status** (p. 40) · **R04 Levofloxacin dose in the rescue regimen** (p. 40) · **R05 Transfusion threshold in acute upper GI bleeding** (p. 40) · **R06 Commonest site of gastric cancer: how common?** (p. 41)

CHAPTER 04 BEFORE MID-MODULE

↑ CONTENTS

Pathology of GIT Tumours

Adds to book pp. 70–75 · book file 1 · book subjects: Pathology · Surgery

IN THIS CHAPTER Carcinoma of the oesophagus · Gastric carcinoma · Adenomas of the colon

Carcinoma of the oesophagus BOOK P. 71

Squamous cell carcinoma under the microscope

PATHOLOGY

It is made of **islands of neoplastic squamous cells**. Differentiated tumours also show **epithelial cell nests**.

E04.01 · Book p. 71 · GIT Others p. 14

A malignant ulcer can bleed fatally

PATHOLOGY

As well as dysphagia and haematemesis, **perforation of the malignant ulcer can cause fatal haemorrhage**.

E04.02 · Book p. 71 · GIT Others p. 14

Gastric carcinoma BOOK PP. 72–73

H. pylori to cancer: the last steps

PATHOLOGY

- ◆ The sequence continues past intestinal metaplasia: H. pylori gastritis → atrophic gastritis → intestinal metaplasia → **dysplasia** → **cancer**. (The atrophic gastritis → intestinal metaplasia step is in Chapter 3 here.)
- ◆ GIT Others lists intestinal metaplasia as a **precancerous (dysplastic) lesion** of the stomach.

E04.03 · Book p. 72 · GIT Others p. 15



Lauren classification: name and frequency of the two types

PATHOLOGY

The intestinal and diffuse types make up the **Lauren classification**: intestinal type **53%**, diffuse type **33%**, unclassified **14%**.

E04.04 · Book p. 72 · GIT Others p. 16

The malignant gastric ulcer

PATHOLOGY

The ulcer is typically **more than 5 cm** across, with an everted edge, a **necrotic floor** and a **firm base**.

E04.05 · Book p. 72 · GIT Others p. 16

Blood spread goes beyond the liver

PATHOLOGY

Blood-borne metastases go to the liver and also to the **lung, bone and brain**.

E04.06 · Book p. 73 · GIT Others p. 16

Adenomas of the colon BOOK P. 73

Colonic adenomas run in families

PATHOLOGY

There is a **well-defined familial predisposition** to colonic adenomas. They predispose to colorectal carcinoma.

E04.07 · Book p. 73 · GIT Others p. 17

Tubular vs villous adenoma: size and cancer risk

PATHOLOGY

	Tubular adenoma	Villous adenoma
Size	Small, <1 cm	Large, >2 cm
Dysplasia / cancer	Rare	High (about 40%)

GIT Others' table gives the 40% for adenomas >4 cm; its text gives it for villous adenomas in general.

E04.08 · Book p. 73 · GIT Others p. 18



Diarrhea and Malabsorption

Adds to book pp. 76–94 · book file 1 · book subjects: Internal Medicine

IN THIS CHAPTER Physiology of fluid and nutrient handling · Mechanisms and causes of diarrhoea · Malabsorption and coeliac disease · Diagnostic tests for malabsorption

Physiology of fluid and nutrient handling BOOK PP. 77–78, 86

Where intestinal water comes from and why it moves

[INTERNAL MEDICINE](#)

- ◆ Water crosses the epithelium **passively**. It follows osmotic gradients created by **active transport** of electrolytes (Na^+ , Cl^-) and of other solutes such as sugars and amino acids.
- ◆ Most of the ~9 L delivered to the small intestine each day is **endogenous secretion**, not oral intake:

Daily fluid delivered to the small intestine (GIT Internal Med p. 49)

Source	Volume (mL)
Oral intake	2,000
Salivary secretion	1,500
Gastric secretion	2,500
Pancreatic secretion	1,500
Biliary secretion	500
Small intestinal secretion	1,000
Total	9,000

E05.01 · Book pp. 77–78 · GIT Internal Med p. 49

Digestive enzymes by source

[INTERNAL MEDICINE](#)

Amylase breaks starch into oligosaccharides, and brush-border enzymes finish the job: lactase splits lactose into glucose + galactose, and sucrase splits sucrose into glucose + fructose.

Digestive enzymes by source (GIT Internal Med p. 50)

Source	Enzymes
Salivary glands	Amylase, lingual lipase
Stomach	Pepsin
Pancreas	Amylase, trypsin, chymotrypsin, carboxypeptidase, elastase, lipase–colipase, phospholipase A2, cholesterol esterase
Intestinal mucosa	Enterokinase; disaccharidases (sucrase, maltase, trehalase, α -dextrinase/isomaltase); peptidases (aminooligopeptidase, dipeptidase)

E05.02 · Book p. 78 · GIT Internal Med p. 50



Where nutrients are absorbed

INTERNAL MEDICINE

The site of disease predicts the deficiency:

- ◆ **Duodenum and jejunum:** calcium, iron, magnesium, phosphorus, folate, the fat-soluble vitamins (A, D, E, K), and lipids, monosaccharides and small peptides.
- ◆ **Ileum: vitamin B12 and bile salts and acids** (also vitamin C, folate, vitamins D and K, magnesium). Many other nutrients can be absorbed here depending on transit time.
- ◆ **Large intestine:** water, sodium, chloride, potassium, short-chain fatty acids, vitamin K and biotin.

E05.03 · Book p. 86 · GIT Internal Med p. 62

Mechanisms and causes of diarrhoea BOOK PP. 79–84

Osmotic and secretory diarrhoea: the mechanism behind the lists

INTERNAL MEDICINE

- ◆ **Osmotic:** usually a poorly absorbable carbohydrate or a **divalent ion** (e.g. magnesium). The high luminal osmolality draws water into the duodenum and jejunum to make chyme isotonic. Na^+ then follows the water into the lumen down its concentration gradient, and this Na^+ pulls in further water.
- ◆ **Secretory laxatives** (examples for the book's "laxatives"): ricinoleic acid (castor oil) and dioctyl sodium sulfosuccinate.

E05.04 · Book pp. 79–80 · GIT Internal Med pp. 52–53

Why oral rehydration works even in cholera

INTERNAL MEDICINE

Even while cholera toxin (via cAMP) drives chloride secretion from the crypts, villus cells can still absorb sodium, and with it water and chloride, through **coupled uptake of sodium with glucose or amino acids**. Oral rehydration solution uses this intact pathway.

E05.05 · Book p. 82 · GIT Internal Med p. 56

Acute infectious diarrhoea: clues from the exposure

INTERNAL MEDICINE

Vehicle and classic pathogens (GIT Internal Med p. 55)

Vehicle	Classic pathogens
Water	Vibrio cholerae, Giardia
Poultry	Salmonella
Seafood and shellfish (e.g. sushi)	Salmonella and Shigella
Canned vegetables or fruits	Clostridium species
Cheese, milk	Listeria species
Hospitalisation, antibiotics or chemotherapy	C. difficile
Animal-to-person (pets)	Salmonella, Campylobacter, Cryptosporidium

The laboratory side of Salmonella and Campylobacter is added in Chapter 7.

E05.06 · Book pp. 81–82 · GIT Internal Med p. 55



Clostridium difficile colitis

INTERNAL MEDICINE

The book names *C. difficile* only as a cause. GIT Internal Med adds:

- ◆ **Pathogenesis:** antibiotics disrupt the normal flora; *C. difficile* colonises the colon and releases **toxins** that cause mucosal damage and inflammation (originally called pseudomembranous colitis).
- ◆ **Diagnosis:** diarrhoea (≥ 3 unformed stools in 24 h) **or** radiographic ileus/toxic megacolon, **plus** a positive stool *C. difficile* toxin test **or** pseudomembranes at colonoscopy/pathology.
- ◆ **Treatment:** stop the offending antibiotic. **Metronidazole** is the drug of choice for mild infection; **vancomycin** for more severe episodes; **fidaxomicin** (a macrocyclic antibiotic) is more active than vancomycin.
- ◆ Probiotics are promising for prevention and treatment. **Faecal microbiota transplantation** is reserved for refractory *C. difficile* colitis.

E05.07 · Book p. 83 · GIT Internal Med pp. 57–58

Chronic diarrhoea: think of the patient group

INTERNAL MEDICINE

Chronic diarrhoea in specific patient groups (GIT Internal Med p. 59)

Patient group	Causes to consider
Travellers	Bacterial infection (mostly acute); protozoal (amoebiasis, giardiasis); tropical sprue
Epidemics and outbreaks	Bacterial infection; epidemic idiopathic secretory diarrhoea (e.g. Brainerd diarrhoea); protozoal (cryptosporidiosis); viral (rotavirus, COVID-19)
Diabetes mellitus	Altered motility; associated coeliac disease, pancreatic exocrine insufficiency, SIBO; drugs (especially acarbose, metformin)
AIDS	Drug side-effects; lymphoma; opportunistic infections (cryptosporidiosis, CMV, HSV, Mycobacterium avium complex)
Institutionalised / hospitalised	<i>C. difficile</i> colitis; drug side-effects; faecal impaction with overflow diarrhoea; ischaemic colitis; tube feeding

E05.08 · Book p. 83 · GIT Internal Med p. 59

Post-infectious IBS: treatable causes to look for

INTERNAL MEDICINE

- ◆ Some cases are due to **bacterial overgrowth** and may respond to non-absorbable antibiotics; others are due to **acquired bile acid malabsorption**.
- ◆ **Coeliac disease** can first appear after an intestinal infection, so screen for it in this setting.

E05.09 · Book p. 84 · GIT Internal Med p. 58

Malabsorption and coeliac disease BOOK PP. 84–92

Maldigestion versus malabsorption

INTERNAL MEDICINE

- ◆ **Maldigestion** is defective intraluminal hydrolysis of nutrients; **malabsorption** is defective mucosal absorption. Their presentations and complications are similar.
- ◆ Many malabsorptive disorders, including coeliac disease, present **subtly** or mainly with **extra-intestinal** features rather than diarrhoea or steatorrhoea. Malabsorption may even be only a laboratory finding.

E05.10 · Book p. 84 · GIT Internal Med p. 60



Coeliac disease: why each extra-intestinal feature occurs

INTERNAL MEDICINE

Selected manifestations and probable causes (GIT Internal Med p. 66)

Manifestation	Probable cause
Dermatitis herpetiformis	Epidermal (type 3) tTG autoimmunity
Ecchymoses, petechiae	Vitamin K deficiency; rarely thrombocytopenia
Oedema	Hypoproteinaemia
Thrombocytosis, Howell–Jolly bodies	Hyposplenism
Secondary hyperparathyroidism	Calcium and/or vitamin D malabsorption with hypocalcaemia
Osteopenia, osteomalacia, osteoporosis	Ca/vitamin D malabsorption, secondary hyperparathyroidism, chronic inflammation
Elevated liver tests	Lymphocytic hepatitis
Peripheral neuropathy	Vitamin B12 and thiamine deficiency; immune-based neurological dysfunction
Ataxia	Cerebellar and posterior column damage
Short stature, delayed puberty, amenorrhoea, infertility	Malnutrition; hypothalamic–pituitary dysfunction
Weakness	Generalised muscle atrophy, hypokalaemia

E05.11 · Book pp. 89–91 · GIT Internal Med p. 66

Coeliac disease: wider associations and later risks

INTERNAL MEDICINE

- ◆ **Further associated conditions** (besides type 1 diabetes and autoimmune liver and thyroid disease): Addison disease, autoimmune haemolytic anaemia, immune thrombocytopenic purpura, **IgA deficiency**, **Down syndrome**, cystic fibrosis, Sjögren syndrome, **microscopic colitis**, epilepsy with cerebral calcification, sarcoidosis, recurrent pericarditis, iridocyclitis/choroiditis, some lung diseases (fibrosing alveolitis, idiopathic pulmonary haemosiderosis, bird-fancier's lung, cavitary lung disease) and macroamylasaemia.
- ◆ **Increased morbidity after diagnosis:** malignancy (non-Hodgkin lymphoma, **small-intestinal adenocarcinoma**); cardiovascular (ischaemic heart disease, cerebrovascular disease, atrial fibrillation, dilated cardiomyopathy); respiratory (COPD, asthma, tuberculosis, influenza).

E05.12 · Book p. 90 · GIT Internal Med p. 66



Diagnostic tests for malabsorption BOOK PP. 93–94

Interpreting the general tests of absorption

INTERNAL MEDICINE

- ◆ **Quantitative stool fat:** a borderline 8–14 g/24 h can occur in secretory or osmotic diarrhoea **not** caused by malabsorption. False negative if fat intake is inadequate; false positive with mineral-oil laxatives or rectal suppositories (e.g. cocoa butter).
- ◆ **Acid steatocrit:** a cheap, easy screening test for fat malabsorption; normal < 10%.
- ◆ **D-xylose test** (used in the book's algorithm, Fig 5.9): tests small-intestinal **mucosal** absorption, so it separates mucosal malabsorption from pancreatic insufficiency. Oral 25 g D-xylose; normal is > 4 g excreted in the urine over 5 h. False positives with renal failure, ascites or incomplete urine collection; it may also be abnormal in bacterial overgrowth and normal in mild or limited mucosal disease.
- ◆ **Lactose H₂ breath test:** a **late** peak (3–6 h) of > 20 ppm suggests lactose malabsorption.
- ◆ **Faecal elastase-1:** as sensitive as the secretin test for moderate-to-severe pancreatic insufficiency, but unreliable in mild insufficiency; false positives with high stool volume and mucosal disease. The **secretin test** is sensitive but labour-intensive and invasive.

E05.13 · Book pp. 93–94 · GIT Internal Med pp. 68–70

Diagnosing bacterial overgrowth

INTERNAL MEDICINE

- ◆ **Gold standard:** quantitative culture of jejunal aspirate; > 10⁵ CFU/mL suggests overgrowth. False negatives occur with focal diverticula or overgrowth distal to the aspiration site.
- ◆ **Glucose H₂ breath test:** an **early** peak (within 2 h) of > 20 ppm suggests overgrowth. Contrast this with the **late** peak of lactose malabsorption. False negatives occur with non-hydrogen-producing organisms.
- ◆ **¹⁴C-D-xylose breath test:** sensitivity and specificity > 90%, and more specific than the H₂ breath test. A non-radioactive ¹³C version is suitable for children and pregnant women.

E05.14 · Book p. 93 · GIT Internal Med pp. 69–70

Biopsy and ileal-function tests

INTERNAL MEDICINE

- ◆ **Small-bowel biopsy:** distal duodenal specimens are usually adequate; take several (4–5). Biopsy is **diagnostic** in intestinal infection, Whipple disease, abetalipoproteinaemia, agammaglobulinaemia, lymphangiectasia, lymphoma and amyloidosis. In coeliac disease and tropical sprue the findings are characteristic, but the diagnosis rests on **improvement after treatment**.
- ◆ **Permeability studies** are gaining favour for screening for small-intestinal disease and monitoring the response to treatment.
- ◆ **SeHCAT:** 7-day retention < 10% is abnormal and indicates bile acid malabsorption; liver disease and bacterial overgrowth can give false results.
- ◆ The **Schilling test** is no longer used.

E05.15 · Book p. 93 · GIT Internal Med p. 69

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R07** How long is "acute" diarrhoea? (p. 41) · **R08** Stool fat and lactose breath test protocols (p. 41)



Constipation

Adds to book pp. 95–105 · book file 1 · book subjects: Internal Medicine

IN THIS CHAPTER Clinical assessment and testing · Disorders of gut–brain interaction and IBS

Clinical assessment and testing BOOK PP. 97–100

History and examination clues

INTERNAL MEDICINE

- ◆ In **acute-onset** constipation, always ask whether the patient can **pass gas**. Failure to pass gas can signal bowel obstruction.
- ◆ **Obstipation** is severe or complete constipation with practically no stool and **no flatus**. It may come with abdominal pain, nausea and vomiting, and points to obstruction.
- ◆ **DRE while bearing down**: pelvic floor dyssynergia is suggested when the anal sphincter **fails to relax** or the perineum **does not descend**.

E06.01 · Book pp. 97–98 · GIT Internal Med p. 71

What anorectal testing measures

INTERNAL MEDICINE

- ◆ **Anorectal manometry**: a pressure-sensitive catheter in the anorectum measures **resting and squeeze pressures** of the anal canal. It assesses the internal and external anal sphincters, **rectal sensation** and expulsion patterns.
- ◆ Defecography can also be performed with **MRI** (MRI defecography).

E06.02 · Book pp. 98, 100 · GIT Internal Med p. 72

Disorders of gut–brain interaction and IBS BOOK PP. 101–105

IBS: extra points

INTERNAL MEDICINE

- ◆ **Functional chest pain of presumed oesophageal origin** is another oesophageal DGBI: chronic, recurrent pain or discomfort referred to the retrosternum.
- ◆ **Autonomic dysfunction** is also listed among the possible contributors to IBS.
- ◆ **History**: fibromyalgia is one of the associations found in IBS patients.
- ◆ **Limited laboratory tests**: besides faecal calprotectin and coeliac serology, GIT Internal Med lists **CBC** (anaemia or infection), **ESR and CRP** (inflammation) and **TSH** (thyroid dysfunction).

E06.03 · Book pp. 101–104 · GIT Internal Med pp. 73–75

Why probiotics are not used in IBS

INTERNAL MEDICINE

Probiotics are not prescribed in IBS because of a **lack of evidence supporting their efficacy**. This is the reason behind the book's "no routine recommendation".

E06.04 · Book p. 105 · GIT Internal Med p. 75

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R09 IBS diagnostic criteria: which Rome edition?** (p. 41) · **R10 Low-FODMAP diet: which foods to limit?** (p. 41) · **R11 IBS-C: the second drug option** (p. 41)



Foodborn Infection

Adds to book pp. 106–114 · book file 1 · book subjects: Microbiology · Internal Medicine

IN THIS CHAPTER Salmonella and enteric fever · Campylobacter and Brucella

Salmonella and enteric fever BOOK PP. 107–110

Salmonella culture and sugar fermentation

MICROBIOLOGY

- ◆ A **facultative anaerobe** that grows well at **37 °C in 24 hours**. Colonies on MacConkey are **pale (yellow)** because it does not ferment lactose.
- ◆ Most species ferment sugars with **acid and gas**. **S. typhi makes acid only** (no gas).

E07.01 · Book p. 107 · GIT Others p. 22

How the first-week blood culture is done

MICROBIOLOGY

Blood is cultured in **fluid (broth) media**, then **subcultured on solid media**, mainly MacConkey, at 37 °C.

E07.02 · Book p. 109 · GIT Others p. 24

Control starts with food handlers

MICROBIOLOGY

Food handlers are examined to detect carriers. Carriers are treated or kept away from handling food. GIT Others lists this before immunisation.

E07.03 · Book p. 109 · GIT Others p. 25

S. typhimurium food poisoning is self-limiting

MICROBIOLOGY

After lasting 2–5 days, **it usually resolves on its own (spontaneous cure)**.

E07.04 · Book p. 110 · GIT Others p. 25

Campylobacter and Brucella BOOK P. 111

Campylobacter laboratory details

MICROBIOLOGY

- ◆ A **rectal swab** is an acceptable specimen, collected with care to avoid oxygen exposure.
- ◆ Darting motility is seen in a wet mount under **dark-field illumination**.
- ◆ Isolated colonies are identified by **biotyping, serotyping and phage typing**.

E07.05 · Book p. 111 · GIT Others p. 27

Brucella needs CO₂

MICROBIOLOGY

Brucella is aerobic, but culture needs **10% CO₂** as well as serum-enriched media.

E07.06 · Book p. 111 · GIT Others p. 27

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R12 Widal test: what a high H titre means** (p. 42)



Inflammatory Bowel Disease

Adds to book pp. 115–121 · book file 1 · book subjects: Internal Medicine · Pharmacology

IN THIS CHAPTER Pathology and diagnosis · Treatment · Complications

Pathology and diagnosis BOOK PP. 116–119

Extra radiological and endoscopic clues

INTERNAL MEDICINE

- ◆ On radiology, "**collar-button**" ulcers are common in UC and only occasional in Crohn's disease.
- ◆ **Colonoscopy (the book's "Endoscopy" heading has no text):**
 - UC: friable mucosa, extensive ulceration and exudates.
 - Crohn's disease: a **single aphthous ulcer** is the earliest finding; multiple oedematous inflammatory polyps give the **cobblestone** appearance.

E08.01 · Book pp. 117–118 · GIT Internal Med pp. 76, 78

More conditions that mimic IBD

INTERNAL MEDICINE

- ◆ Other colitides: **radiation colitis**, **Behçet's colitis**, **schistosomiasis**, and sexually transmitted colitis (gonococcus, *Chlamydia*, herpes, trauma).
- ◆ Bacterial colitis also includes **Shigella** and **invasive E. coli**.
- ◆ Crohn's look-alikes: **lymphoma**, **Yersinia** and **tuberculosis**.

E08.02 · Book p. 119 · GIT Internal Med p. 79

Treatment BOOK PP. 119–120

The missing drug groups and a severity ladder

INTERNAL MEDICINE

GIT Internal Med lists six drug groups for IBD. Two of them, **5-aminosalicylates (5-ASA/sulfasalazine)** and **probiotics/prebiotics**, do not appear in the book's IBD chapter. It also steps treatment up with severity:

Stepwise IBD therapy (GIT Internal Med p. 81)

Severity	Treatment
Mild	5-aminosalicylates / sulfasalazine; local steroid enemas; local 5-ASA enemas and suppositories
Moderate	Prednisolone; budesonide; immunosuppressant
Severe	Biological therapy; surgery

E08.03 · Book pp. 119–120 · GIT Internal Med pp. 80–81



5-ASA and budesonide: the formulation decides where the drug acts

INTERNAL MEDICINE

Choose the formulation by disease site. Most act in the colon; Pentasa is released from the duodenum onwards; enemas reach the rectum and sigmoid, and suppositories the rectum.

5-aminosalicylates and budesonide (GIT Internal Med p. 80 drug table)

Drug	Dose	Release site
Sulfasalazine (Azulfidine)	2–6 g/day	Colon
Mesalamine (Asacol, Lialda, Apriso)	2.4–4.8 g/day	Distal ileum, colon
Olsalazine (Dipentum)	1–3 g/day	Colon
Balsalazide (Colazal)	6.25 g/day	Colon
Mesalamine (Pentasa)	2–4 g/day	Duodenum, jejunum, ileum, colon
Mesalamine (Rowasa) enema; suppository	4 g/day (enema); 1 g/day (suppository)	Rectum/sigmoid; rectum
Mesalamine (Canasa) suppository	1 g/day	Rectum
Budesonide (Entocort EC)	Induction 9 mg PO daily; maintenance 6 mg PO daily	Small intestine
Budesonide (MMX, UCERIS)	Induction 9 mg PO daily	Colon

The systemic-steroid rows of the same table disagree with the book and with the source's own text, so they are kept for review.

E08.04 · Book pp. 119–120 · GIT Internal Med p. 80 (drug table)

Immunomodulators and biologics: full list and doses

INTERNAL MEDICINE

The book names azathioprine, methotrexate, cyclosporin, infliximab and vedolizumab without doses. GIT Internal Med adds **6-mercaptopurine** and four more biologics: **adalimumab, golimumab, certolizumab pegol, natalizumab**.

Immunomodulators and biologics (GIT Internal Med p. 80 drug table)

Drug	Induction / dose	Maintenance
6-Mercaptopurine	1.5 mg/kg/day	–
Azathioprine	2.5 mg/kg/day	–
Methotrexate	25 mg SC weekly × 4 months	15–25 mg SC weekly
Cyclosporine	2–4 mg/kg/day IV	–
Infliximab	5 mg/kg IV weeks 0, 2, 6	5–10 mg/kg IV every 8 weeks
Adalimumab	160 mg SC week 0, 80 mg week 2	40 mg SC every other week
Golimumab	200 mg SC week 0, 100 mg week 2	100 mg SC every 4 weeks
Certolizumab pegol	400 mg SC weeks 0, 2, 4	400 mg SC every 4 weeks
Natalizumab	300 mg IV every 4 weeks	–
Vedolizumab	300 mg IV at 0, 2 and 6 weeks	300 mg IV every 8 weeks

E08.05 · Book p. 120 · GIT Internal Med p. 80 (drug table)



Complications BOOK P. 121

Complications grouped by disease

INTERNAL MEDICINE

- ◆ **UC:** toxic megacolon and malignancy.
- ◆ **Crohn's disease:** abscesses and fistulas, obstruction, and perianal disease.

E08.06 · Book p. 121 · GIT Internal Med p. 82

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R13 Ulcerative colitis: two rows of GIT Internal Med's p. 76 table vs the book's Fig 8.1** (p. 42) · **R14 Systemic steroid doses in IBD** (p. 42)

CHAPTER 09

Parasitic GI Diseases NO ADDITIONS

Book pp. 122–139 · book file 1 · Parasitology

GIT Others pp. 35–47 repeat this chapter (pp. 122–139) word for word, with the same doses and time periods; GIT Internal Med has no parasitology section.

CHAPTER 10

Small Intestine and Colon Surgery NO ADDITIONS

Book pp. 140–144 · book file 1 · Surgery

Neither GIT Internal Med nor GIT Others adds anything to this chapter (Meckel's diverticulum and small-bowel/colon surgery).

CHAPTER 11

Diverticulosis of the Colon NO ADDITIONS

Book pp. 145–150 · book file 1 · Surgery · Internal Medicine

GIT Internal Med mentions diverticular disease only as a cause of lower GI bleeding and left iliac fossa pain, already in the book or in Chapter 3 here.

CHAPTER 12

Appendix NO ADDITIONS

Book pp. 151–161 · book file 1 · Surgery

Appendicitis appears in GIT Internal Med only in its list of right iliac fossa pain causes, which the book already gives.

CHAPTER 13

Intestinal Obstruction NO ADDITIONS

Book pp. 162–188 · book file 1 · Surgery · Emergency / Acute Care

GIT Internal Med's imaging points on obstruction (plain films, CT in high-grade obstruction, enteroclysis) are in Chapter 1 here, where the book teaches imaging.

CHAPTER 14 BEFORE MID-MODULE

↑ CONTENTS

Polyposis of Colon

Adds to book pp. 189–194 · book file 1 · book subjects: Surgery · Pathology

Familial polyposis coli BOOK P. 190

Polyp type in FAP

PATHOLOGY

In FAP, **most of the polyps are tubular adenomas.**

E14.01 · Book p. 190 · GIT Others p. 19



Colorectal Cancer

Adds to book pp. 195–209 · book file 1 (pp. 195–197) and file 2 (pp. 197–209) · book subjects: Surgery · Pathology

Pathology of colorectal carcinoma BOOK P. 196

Adenocarcinoma: 98% of large-bowel cancers

PATHOLOGY

The book says "more than 90%" of colorectal carcinomas are adenocarcinomas. GIT Others gives a precise figure: **98%** of all cancers of the large intestine are adenocarcinomas.

E15.01 · Book p. 196 · GIT Others p. 19

Two kinds of mucoid (mucinous) carcinoma

PATHOLOGY

- ◆ **Mucinous adenocarcinoma:** abundant *extracellular* mucin (the book: >50% mucinous material).
- ◆ **Signet ring cell carcinoma:** the mucin stays *inside* the malignant cell and pushes the nucleus to one side (eccentric nucleus).

E15.02 · Book p. 196 · GIT Others p. 20

Add irradiation to the predisposing factors

PATHOLOGY

GIT Others lists **irradiation** as a predisposing factor for colorectal adenocarcinoma, along with adenoma, familial polyposis, IBD and gene mutations. It is not in the book's list.

E15.03 · Book p. 196 · GIT Others p. 19

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R15 Colorectal cancer: age at presentation** (p. 42)



PART 2 · AFTER MID-MODULE

Chapters 16–40

Book file 2, printed pp. 197–433 (Chapter 16 starts on p. 210). Most additions here come from the Hepatology source.

CHAPTER 16

Anorectal Diseases

NO ADDITIONS

Book pp. 210–230 · book file 2 · Surgery

GIT Others was checked against this chapter and adds nothing new; the Hepatology source does not cover anorectal disease.

CHAPTER 17

AFTER MID-MODULE

↑ CONTENTS

Hepatology and Hepatic Surgery

Adds to book pp. 231–235 · book file 2 · book subjects: Surgery

Liver functions and lobular zonation

BOOK PP. 233–235

Principal liver functions

HEPATOLOGY

Bile formation and secretion; metabolism of carbohydrates, amino acids, lipids and water- and fat-soluble vitamins; inactivation of ammonia (urea cycle), toxins, steroids and other hormones; synthesis of acute-phase proteins, albumin, clotting factors and steroid/hormone-binding proteins; immune function (Kupffer cells).

E17.01 · Book p. 233 · Hepatology p. 64

Metabolic zonation of the lobule

HEPATOLOGY

The **periportal** zone carries out gluconeogenesis and cholesterol and urea synthesis; the **pericentral** zone carries out glycolysis and glutamine and bile-acid synthesis.

E17.02 · Book p. 235 · Hepatology p. 64 (figure)

CHAPTER 18

AFTER MID-MODULE

↑ CONTENTS

Liver Disease Assessment

Adds to book pp. 236–250 · book file 2 · book subjects: Radiology / Investigation · Internal Medicine

IN THIS CHAPTER Symptoms and signs of chronic liver disease · Liver function tests · Bilirubin and bile-acid metabolism · Types of jaundice · Viral hepatitis markers

Symptoms and signs of chronic liver disease

BOOK PP. 237–238

Palmar erythema and spider naevi: what they are

HEPATOLOGY

Palmar erythema is persistent redness of the palms; spider naevi are small red skin marks caused by dilated capillaries near the skin surface.

E18.01 · Book p. 238 · Hepatology p. 1



Hyperdynamic circulation and its vasodilators

HEPATOLOGY

- ◆ **Signs:** flushed extremities, bounding pulses and capillary pulsations, with increased portal blood flow, low blood pressure and reduced systemic vascular resistance; cardiac output rises.
- ◆ **The vasodilators are probably multiple:** formed by the sick hepatocyte, not inactivated by it, or bypassing it through intra- or extra-hepatic portosystemic shunts. They activate the sympathetic and renin-angiotensin systems, promoting sodium and water retention and ascites.
- ◆ **Nitric oxide** (endothelium-derived, potent vasodilator) is implicated in the hyperdynamic circulation and is important in ascites, hepatorenal syndrome and portal hypertension. Prostanoids released into the portal vein and endocannabinoids are also shown as splanchnic vasodilators.

E18.02 · Book p. 237 · Hepatology p. 2 (text + figure)

Hepatocellular failure: when it occurs

HEPATOLOGY

Hepatocellular failure complicates almost all liver diseases but **not portal venous occlusion alone**; it may be the terminal event in chronic cholestasis such as primary biliary cirrhosis. Its picture and treatment are similar whatever the cause.

E18.03 · Book pp. 237–238 · Hepatology p. 64

Liver function tests BOOK PP. 238, 244–246

What the liver makes, and what albumin and PT tell you

HEPATOLOGY

- ◆ The liver synthesises albumin, **all clotting factors except factor VIII**, carrier proteins (ceruloplasmin, transferrin), cholesterol, bile salts and urea.
- ◆ The source links the bleeding tendency of chronic liver disease to failed synthesis of fibrinogen and factors II, V, VII, IX, XI, XII and XIII.
- ◆ A falling serum albumin is an excellent indicator of severity and prognosis in chronic liver disease. PT depends on factors I, II, V, VII and X; because these have a shorter half-life than albumin, PT is an **earlier** indicator of severe liver injury.

E18.04 · Book pp. 238, 244, 246 · Hepatology pp. 1, 5, 10

Liver tests grouped by what they reflect

HEPATOLOGY

- ◆ **Excretory / detoxification:** serum and urine bilirubin, serum ammonia.
- ◆ **Liver-cell injury:** AST, ALT.
- ◆ **Cholestasis:** ALP, GGT, 5'-nucleotidase. 5'-nucleotidase is more specific for cholestasis than ALP but is not widely used.
- ◆ **Synthetic function:** albumin, coagulation factors.

E18.05 · Book p. 245 · Hepatology pp. 6, 87 (table)



ALP and GGT with space-occupying or infiltrative lesions

HEPATOLOGY

- ◆ ALP and GGT are markedly raised (>3-fold) not only in cholestasis but also with **space-occupying lesions and hepatic infiltration** (tumours such as cancer, cysts, lymph nodes). Non-hepatic ALP rises occur in bone disease (rickets, osteomalacia).
- ◆ In infiltration, bilirubin may be **normal**: partial obstruction lets bilirubin pass yet stimulates ALP/GGT synthesis. AST/ALT are normal or slightly raised (AST > ALT). AFP rises in HCC, and CEA with secondaries from GIT cancer.

E18.06 · Book p. 245 · Hepatology pp. 6, 20

Causes of an isolated ALP rise

HEPATOLOGY

Primary or metastatic hepatic tumours, granulomatous liver disease, primary biliary cirrhosis, non-Hodgkin lymphoma, hyperthyroidism, bone disease with rapid turnover, sclerosing cholangitis.

E18.07 · Book p. 245 · Hepatology p. 87 (table)

Bilirubin and bile-acid metabolism BOOK P. 245

Where bilirubin comes from

HEPATOLOGY

About 250–300 mg of bilirubin is produced daily; 70–80% comes from haemoglobin of senescent red cells, the rest from prematurely destroyed erythroid cells in the marrow and turnover of haemoproteins such as myoglobin. It is formed in reticuloendothelial cells, mainly in the spleen and liver.

E18.08 · Book p. 245 · Hepatology p. 81

Hepatocyte handling in four steps

HEPATOLOGY

Uptake, cytosolic binding, conjugation, secretion: albumin-bound bilirubin is taken up (albumin is not), binds **ligandin (Z protein)**, is conjugated to glucuronic acid by UDP-glucuronosyl transferase in the endoplasmic reticulum, and mono- and diglucuronides are actively transported into bile.

E18.09 · Book p. 245 · Hepatology pp. 7, 81–82

In the gut

HEPATOLOGY

- ◆ Conjugated bilirubin is not absorbed by the intestinal mucosa; only in the distal ileum and colon is it hydrolysed (bacterial β -glucuronidases) to unconjugated bilirubin, then reduced to colourless urobilinogens.
- ◆ The primary bile acids, cholic and chenodeoxycholic acid, are made from cholesterol, secreted in bile, and mostly reabsorbed in the ileum for re-secretion (enterohepatic circulation).

E18.10 · Book p. 245 · Hepatology pp. 7, 82



Types of jaundice BOOK PP. 245–246

Haemolytic, obstructive and hepatic jaundice: details the book's comparison lacks

HEPATOLOGY

- ◆ Urine **bile salts** (with bilirubin) are absent in haemolytic jaundice and present in obstructive and hepatic jaundice. In hepatic jaundice the stool is **clay-coloured**, as in obstruction.
- ◆ In haemolytic jaundice, evidence of haemolysis includes raised reticulocytes **and raised LDH** (the LDH–haptoglobin–reticulocyte algorithm is in Chapter 29). Obstructive jaundice raises serum cholesterol and **β-lipoproteins**.

E18.11 · Book pp. 245–246 · Hepatology pp. 9–10 (table)

Crigler–Najjar II, Dubin–Johnson and Rotor

HEPATOLOGY

- ◆ Crigler–Najjar **type II** = partial UGT deficiency, bilirubin <20 mg/dL (type I: complete absence, >20 mg/dL).
- ◆ Dubin–Johnson (defect of bilirubin secretion) has a **darkly pigmented** liver; Rotor syndrome is usually asymptomatic with a non-pigmented liver.

E18.12 · Book pp. 245–246 · Hepatology p. 8

Viral hepatitis markers BOOK PP. 247–250

Reading HBV serology

HEPATOLOGY

HBV serology panels

HBsAg	Anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
–	–	–	–	Susceptible
+	+	+	–	Acutely infected
+	+	–	–	Chronically infected
–	+	–	+	Immune: natural infection
–	–	–	+	Immune: vaccination

- ◆ **Acute vs chronic vs past HBV:** HBeAg and HBV DNA are + early then – in acute infection, ± in chronic, – in past; anti-HBe is – early then + (acute), ± (chronic), + (past); ALT is markedly raised (acute), mildly-moderately raised (chronic), normal (past).
- ◆ An **isolated anti-HBc** in the window period usually means a resolving acute infection that is **not yet completely resolved**, because anti-HBs has not yet appeared.

E18.13 · Book p. 248 · Hepatology p. 17 (tables)

Normal ALT/AST does not prove recovery

HEPATOLOGY

In hepatitis B and C, normal ALT/AST does not prove recovery: replication may persist, so clearance must be shown serologically.

E18.14 · Book p. 247 · Hepatology p. 14

HAV RNA in faeces

HEPATOLOGY

HAV can be detected by **PCR on faeces**, earlier than serology, but this is very rarely performed.

E18.15 · Book p. 247 · Hepatology p. 15



Anti-HCV pitfalls

HEPATOLOGY

A positive anti-HCV may be false (cross-reactivity with another infection) and must be confirmed by PCR; a negative result does not exclude HCV (immunodeficiency; very early infection before seroconversion: do PCR). **Anti-HCV IgM is not a marker of acute infection.**

E18.16 · Book pp. 248–249 · Hepatology p. 18

Hepatitis G

HEPATOLOGY

Hepatitis G is spread by contaminated blood or blood products; prevalence in blood donors is about 1%.

E18.17 · Book p. 249 · Hepatology p. 20

Chronic hepatitis at a glance

HEPATOLOGY

	Autoimmune	HBV	HCV
Sex	Female	Male	Equal
Age	15–25 y	Older	All ages
Associated autoimmune disease	Frequent	Rare	Occasional
Autoantibodies	High titre (ANA, ASMA, LKM)	Low / absent	Low / absent
Cancer risk	Low	High	High

E18.18 · Book p. 250 · Hepatology p. 21 (table)

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R16 PBC in the immunoglobulin list: raised AMA or raised IgM?** (p. 42) · **R17 What happens to the reabsorbed 10–20% of urobilinogen?** (p. 42) · **R18 Gilbert's syndrome: decreased uptake or decreased conjugation?** (p. 43)

CHAPTER 19 AFTER MID-MODULE

↑ CONTENTS

Acute Hepatitis

Adds to book pp. 251–263 · book file 2 · book subjects: Internal Medicine

IN THIS CHAPTER Course and prognosis · Virology and transmission · Chronic hepatitis

Course and prognosis BOOK PP. 252, 257

How long acute viral hepatitis lasts

HEPATOLOGY

Acute viral hepatitis usually resolves 4–8 weeks after symptom onset.

E19.01 · Book p. 257 · Hepatology p. 14

PT, not ALT, judges severity

HEPATOLOGY

Severity is judged by impaired synthetic function rather than ALT/AST levels: **PT** is the key prognostic indicator, and a PT more than 3 seconds above normal carries a poor prognosis.

E19.02 · Book p. 252 · Hepatology p. 14

Virology and transmission BOOK PP. 254–256

HBV is in every body fluid except stool

HEPATOLOGY

HBV is present in all body fluids **except stool**, so spread is by blood or body-fluid contact.

E19.03 · Book p. 255 · Hepatology p. 16



HCV and transfusion

HEPATOLOGY

HCV (enveloped ssRNA flavivirus) is the most common cause of transfusion-associated hepatitis; sexual transmission is low.

E19.04 · Book p. 256 · Hepatology p. 18

Why HDV needs HBV

HEPATOLOGY

HDV causes infection only when it is **encapsulated with HBsAg**, which is the basis of its dependence on HBV.

E19.05 · Book p. 255 · Hepatology p. 19

HEV genotypes and chronic HEV

HEPATOLOGY

Genotypes 1 and 2 cause waterborne outbreaks; genotypes 3 and 4 cause sporadic cases, including via undercooked meat such as pork. HEV affects mainly adults and is rare in children. Chronic (genotype 3) hepatitis E occurs only in the immunocompromised: organ-transplant recipients, cancer chemotherapy, HIV.

E19.06 · Book pp. 254, 258 · Hepatology p. 19

Chronic hepatitis BOOK P. 263

Rosettes in severe chronic hepatitis

HEPATOLOGY

In severe chronic hepatitis, fibrous septa extend into the lobule, producing **rosette formation** as well as bridging fibrosis.

E19.07 · Book p. 263 · Hepatology p. 34

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R19 Rest in acute viral hepatitis** (p. 43)

CHAPTER 20 AFTER MID-MODULE

↑ CONTENTS

Chronic Hepatitis B

Adds to book pp. 264–273 · book file 2 · book subjects: Internal Medicine · Pharmacology

IN THIS CHAPTER Natural history of HBV · Treatment of HBV · Hepatitis C (within this chapter)

Natural history of HBV BOOK PP. 265–267

Who becomes a carrier, and what follows

HEPATOLOGY

Chronic HBV is more common in males. About 50% of HBsAg carriers have evidence of replication (HBeAg / HBV DNA positive), and about 20% of carriers develop cirrhosis within 5 years of disease onset.

E20.01 · Book p. 265 · Hepatology p. 35

Loss of immune tolerance

HEPATOLOGY

Immune-tolerant patients (mostly infected perinatally) often lose tolerance in the 3rd–5th decade, when acute hepatitis develops and chronic hepatitis may follow.

E20.02 · Book p. 266 · Hepatology p. 35



Inactive carriers and reactivation

HEPATOLOGY

Inactive carriers are usually discovered on screening before blood donation. Reactivation can occur with cancer chemotherapy or bone-marrow transplantation, so **antiviral prophylaxis is indicated** in these settings.

E20.03 · Book p. 267 · Hepatology p. 35

Treatment of HBV BOOK PP. 269–271

HBeAg loss

HEPATOLOGY

When HBeAg disappears, remission is usually sustained for many years.

E20.04 · Book p. 269 · Hepatology p. 36

Pegylated interferon: who responds

HEPATOLOGY

Pegylated interferon α -2a for 48 weeks gives response rates of about 25–45%; response is best in younger patients with ALT about 3× the upper limit of normal.

E20.05 · Book pp. 270–271 · Hepatology p. 36

Entecavir, tenofovir and lamivudine

HEPATOLOGY

- ◆ **Entecavir:** HBV DNA becomes negative by 48 weeks in 67% (HBeAg-positive) and 90% (HBeAg-negative).
- ◆ **Tenofovir** can be used for lamivudine mutations but not alone for adefovir mutations.
- ◆ **Lamivudine** (100 mg/day) is well tolerated but limited by resistance from the **YMDD mutant**.

E20.06 · Book p. 270 · Hepatology p. 37

Hepatitis C (within this chapter) BOOK PP. 271–273

How DAAs work

HEPATOLOGY

DAAs target specific HCV **non-structural (NS) proteins** vital for viral replication, disrupting replication and infection.

E20.07 · Book p. 272 · Hepatology p. 38

Sofosbuvir and ribavirin: dose and adverse effects

HEPATOLOGY

- ◆ **Sofosbuvir:** 400 mg once daily with or without food; adverse effects (~10%) fatigue, headache, anorexia, insomnia.
- ◆ **Ribavirin** (adjuvant): 600–1200 mg/day by weight; adverse effects haemolytic anaemia, teratogenicity, sperm abnormalities.

E20.08 · Book pp. 272–273 · Hepatology p. 38

Why Egypt's HCV burden is so high

HEPATOLOGY

Egypt's HCV burden also reflects nosocomial spread through inadequately disinfected equipment (e.g. circumcision, dental practice), besides the antischistosomal injection campaigns.

E20.09 · Book p. 273 · Hepatology p. 39

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R20 HBV prevalence in Egypt** (p. 43) · **R21 Risk of chronic HBV by age at infection** (p. 43) · **R22 Inactive carrier: HBV DNA level** (p. 43) · **R23 Spontaneous HBsAg clearance per year** (p. 43) · **R24 Egypt's HCV screening campaign: how many screened?** (p. 43)



CHAPTER 21 AFTER MID-MODULE

[↑ CONTENTS](#)

Autoimmune Hepatitis

Adds to book pp. 274–279 · book file 2 · book subjects: Internal Medicine

IN THIS CHAPTER AIH subtypes · Fatty liver (MASLD) management

AIH subtypes BOOK P. 275**Type 1 vs type 2 AIH**

HEPATOLOGY

	Type 1	Type 2
Autoantibodies	ANA alone 10% · ASMA alone 35% · both 50% · neither 5%	LKM-1, LC-1
Serum IgG	+++	+
Usual age	10 years to elderly	2–18 years
Progression to cirrhosis	≈45%	≈80%

How common each type is has been left out: the source and the book give different figures (review point R25).

E21.01 · Book p. 275 · Hepatology p. 22 (table)

Fatty liver (MASLD) management BOOK P. 279**Weight loss should be gradual**

HEPATOLOGY

Weight loss should be gradual, **not more than 1.6 kg per week**, because rapid weight loss increases fibrosis.

E21.02 · Book p. 279 · Hepatology p. 44

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R25 How common is type 1 autoimmune hepatitis?** (p. 44) · **R26 Metformin in fatty liver disease** (p. 44)

CHAPTER 22 AFTER MID-MODULE

[↑ CONTENTS](#)

Liver Cirrhosis

Adds to book pp. 280–307 · book file 2 · book subjects: Internal Medicine · Pathology

IN THIS CHAPTER Aetiology and consequences · Clinical picture and investigations · Varices screening without endoscopy · Cholestatic and metabolic cirrhosis

Aetiology and consequences BOOK PP. 281–282**Cryptogenic cirrhosis**

HEPATOLOGY

Some cirrhosis is **cryptogenic** (no identifiable cause).

E22.01 · Book p. 281 · Hepatology p. 65



Two routes by which cirrhosis causes disease

HEPATOLOGY

- ◆ **Altered hepatic blood flow:** portal hypertension → variceal bleeding, ascites, encephalopathy; ascites → SBP, HRS.
- ◆ **Reduced functional cell mass:** less detoxification of bilirubin, ammonia and drugs; less albumin and clotting-protein synthesis.

E22.02 · Book p. 282 · Hepatology p. 65 (figure)

Clinical picture and investigations BOOK PP. 282–284

Associations, fever and feminisation

HEPATOLOGY

Recurrent pancreatitis (especially in alcoholics) is among the associated conditions. Fever may be due to portal bacteraemia or liver-cell necrosis, not only infection. Feminisation in males is seen in long-standing cases and usually indicates **decompensation**.

E22.03 · Book p. 282 · Hepatology p. 45

Urine urobilinogen

HEPATOLOGY

Urine urobilinogen is increased in cirrhosis.

E22.04 · Book p. 284 · Hepatology p. 47

Varices screening without endoscopy BOOK P. 285

Elastography cut-offs

HEPATOLOGY

- ◆ Liver stiffness ≤ 20 kPa with platelets $>150 \times 10^9/L$: screening endoscopy not needed (large varices <5%).
- ◆ Liver stiffness >25 kPa is sufficient to diagnose clinically significant portal hypertension (CSPH), with no endoscopy needed; treat with carvedilol to reduce the risk of decompensation.

E22.05 · Book p. 285 · Hepatology pp. 48, 60

Cholestatic and metabolic cirrhosis BOOK PP. 287–300

PBC: supplements and biopsy

HEPATOLOGY

Supplement fat-soluble vitamins and calcium. Biopsy shows loss of small bile ducts with **ductular proliferation**.

E22.06 · Book pp. 287–288 · Hepatology p. 51

Haemochromatosis: skin, cancer risk and MRI

HEPATOLOGY

- ◆ Bronze skin is due to **melanin** deposition; about 30% may develop HCC as a complication of cirrhosis.
- ◆ MRI showing iron in the **pancreas** distinguishes hereditary haemochromatosis from secondary iron overload (haemosiderosis), where only the liver is affected.

E22.07 · Book pp. 292–293 · Hepatology p. 53



Wilson disease: copper handling and features

HEPATOLOGY

- ◆ Ceruloplasmin is an α_2 -globulin carrying six copper atoms (loaded by ATP7B); 50–75% of intestinal copper is absorbed; normal serum ceruloplasmin is 20–35 mg/dL.
- ◆ Haemolysis is due to the toxic effect of copper on red cells; psychiatric disease can extend to psychosis (schizophrenia-like or manic-depressive).
- ◆ Kayser–Fleischer rings are gold or greenish-gold rings at the corneoscleral junction; in a few people they are the **first sign** of Wilson disease.

E22.08 · Book pp. 294–296 · Hepatology pp. 12–13

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R27 Feter hepaticus: what it smells like** (p. 44) · **R28 Dietary protein in hepatic encephalopathy** (p. 44) · **R29 PBC: how many diagnostic criteria are needed?** (p. 44) · **R30 UDCA in PBC: does it change the disease?** (p. 44) · **R31 Alcohol threshold for alcoholic cirrhosis: grams or millilitres?** (p. 44)

CHAPTER 23 AFTER MID-MODULE

↑ CONTENTS

Portal Hypertension

Adds to book pp. 308–317 · book file 2 · book subjects: Internal Medicine · Surgery

IN THIS CHAPTER Causes and splenomegaly · Primary prophylaxis · Acute variceal bleeding

Causes and splenomegaly BOOK PP. 309–311

Causes by site that the book does not list

HEPATOLOGY

Site	Additional causes in the source
Increased flow	Idiopathic tropical splenomegaly; arteriovenous fistula
Prehepatic	Splenic-vein thrombosis; invasion or compression of portal/splenic vein by tumour
Presinusoidal	Early PBC; chronic active hepatitis; vinyl chloride, copper; idiopathic portal hypertension
Sinusoidal (non-cirrhotic)	Cytotoxic drugs; vitamin A intoxication
Post-sinusoidal	Alcoholic central hyaline sclerosis
Posthepatic	IVC webs, tumour invasion or thrombosis

E23.01 · Book pp. 309–310 · Hepatology p. 57 (figure)

Why the spleen enlarges

HEPATOLOGY

Splenomegaly results from congestion **and** reticulo-endothelial hyperplasia.

E23.02 · Book p. 311 · Hepatology p. 59

Primary prophylaxis BOOK P. 313

Variceal haemorrhage as a cause of death

HEPATOLOGY

Variceal haemorrhage accounts for about **one-third** of deaths in cirrhosis with portal hypertension.

E23.03 · Book p. 313 · Hepatology p. 60



How non-selective β -blockers work

HEPATOLOGY

Non-selective β -blockers reduce portal inflow by lowering cardiac output (β_1) and blocking β_2 vasodilator receptors on splanchnic arteries; they are **as effective as banding**.
 Propranolol alternative: 80–160 mg/day, titrated to cut resting pulse by 25%, for life.

E23.04 · Book p. 313 · Hepatology p. 61

Acute variceal bleeding BOOK PP. 314–317

Blood-pressure target and terlipressin regimen

HEPATOLOGY

- ◆ Resuscitate to a systolic BP of about 100 mmHg.
- ◆ Terlipressin: 2 mg IV every 6 h, then 1 mg every 4 h, for 2–5 days. The book gives the duration (2–5 days) but no dose; this dose is from the Hepatology source only. (That source's next sentence on ischaemic heart disease is garbled; see note N4.)

E23.05 · Book p. 315 · Hepatology pp. 61–62

Histoacryl and balloon tamponade

HEPATOLOGY

Histoacryl (tissue adhesive) is used for gastric varices **and big oesophageal varices**.
 Balloon tamponade is rarely used now because of its complications, but it can be used when endoscopic or vasoconstrictor therapy has failed or is unavailable.

E23.06 · Book pp. 315–316 · Hepatology p. 62

TIPS timing and surgical shunts

HEPATOLOGY

- ◆ The source gives both steps together: a rebleed within 10 days is treated by repeating endoscopic therapy (as in the book, p. 316), and **TIPS** is used when bleeding cannot be stopped or early rebleeding occurs **within 5 days** of endoscopic therapy.
- ◆ Surgical shunts are usually end-to-side portocaval or Warren (splenorenal).

E23.07 · Book pp. 316–317 · Hepatology pp. 62–63

FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R32 Transfusion target in acute variceal bleeding** (p. 45) · **R33 Nasogastric (Ryle's) tube in GI bleeding: routine or selective?** (p. 45) · **R34 Screening endoscopy for every cirrhotic?** (p. 45)

CHAPTER 24

Hydatid Cyst

NO ADDITIONS

Book pp. 318–323 · book file 2 · Parasitology · Surgery

GIT Others names hydatid disease only as a differential of amoebic liver abscess, which the book already covers (Ch 9); the Hepatology source does not cover hydatid cyst.

CHAPTER 25

Pyogenic Liver Abscess

NO ADDITIONS

Book pp. 324–331 · book file 2 · Surgery · Internal Medicine · Parasitology

The one finding for this chapter is a disagreement about the blood count in amoebic liver abscess, kept for review. Review point: **R35 Blood count in amoebic liver abscess** (p. 45).

CHAPTER 26

Hepatocellular Carcinoma

NO ADDITIONS

Book pp. 332–336 · book file 2 · Surgery · Pathology

Neither GIT Others nor the Hepatology source adds anything to hepatocellular carcinoma beyond the book.



Ascites

Adds to book pp. 337–344 · book file 2 · book subjects: Internal Medicine

IN THIS CHAPTER Causes and mechanism · Investigations · Management · Spontaneous bacterial peritonitis

Causes and mechanism BOOK PP. 338–339

Commonest causes after cirrhosis

HEPATOLOGY

After cirrhosis, malignancy is the next commonest cause; cardiac failure and tuberculous peritonitis are less frequent.

E27.01 · Book p. 338 · Hepatology p. 66

Causes grouped by portal hypertension

HEPATOLOGY

Causes	
With PHT: intrahepatic	Cirrhosis, alcoholic hepatitis, active chronic hepatitis, liver failure, veno-occlusive disease, liver tumour, schistosomiasis, sarcoidosis, Budd–Chiari, focal nodular hyperplasia
With PHT: extrahepatic	Heart failure, pericarditis, portal-vein thrombosis
No PHT: peritoneal	Carcinomatosis, mesothelioma, pseudomyxoma, infection
No PHT: hypoalbuminaemia	Nephrotic syndrome, malnutrition, bowel disease
No PHT: mixed	Myxoedema, pancreatic/biliary ascites, amyloidosis, familial Mediterranean fever, granulomatous disease

E27.02 · Book p. 338 · Hepatology p. 66

RAAS: one activation, two consequences

HEPATOLOGY

Renin (juxtaglomerular cells) converts liver-made angiotensinogen to angiotensin I; ACE (mainly pulmonary endothelium) forms angiotensin II, a pressor that also releases aldosterone. Release is driven by renal perfusion (JG cells), distal Na/Cl load (macula densa) and sympathetic β -receptors. The same RAAS activation causes **renal sodium avidity (ascites)** and **renal vasoconstriction (HRS)**.

E27.03 · Book p. 339 · Hepatology pp. 67–68

Investigations BOOK P. 340

Detecting and tapping ascites

HEPATOLOGY

- ◆ Ultrasound detects as little as 30 mL (right lateral decubitus).
- ◆ Diagnostic paracentesis is indicated for new-onset ascites or a suspected complication (sterile, 20–23-gauge angiocatheter).
- ◆ Extra fluid tests: Gram stain, triglycerides (chylous), bilirubin (bowel perforation), glucose, amylase, LDH, cytology. Pair ascitic with same-day serum albumin for SAAG.

E27.04 · Book p. 340 · Hepatology p. 70



SAAG: accuracy and extra causes

HEPATOLOGY

SAAG correlates directly with portal pressure; ≥ 1.1 g/dL (11 g/L) vs < 1.1 separates portal-hypertensive ascites with $> 97\%$ accuracy.

- ◆ **Extra high-gradient causes:** fatty liver of pregnancy, massive metastases, mixed ascites, myxoedema, portal-vein thrombosis, sinusoidal obstruction syndrome.
- ◆ **Extra low-gradient causes:** bowel obstruction/infarction, postoperative lymphatic leak, serositis in connective-tissue disease.

E27.05 · Book p. 340 · Hepatology p. 71 (text + table)

Management BOOK PP. 341–342

Diuretics: sodium intake, timing and when to stop

HEPATOLOGY

The book already gives bed rest, daily weight and spironolactone 100–200 mg. The source adds a **70–90 mmol sodium diet; after 4 days** consider adding furosemide 40 mg daily; and **stop diuretics** if pre-coma (flap), hypokalaemia, azotaemia or alkalosis.

E27.06 · Book p. 341 · Hepatology p. 72 (box)

Refractory ascites: two kinds, and what drives it

HEPATOLOGY

Refractory ascites (10–20% of patients) is diuretic-**resistant** (no response to maximal doses) or diuretic-**intractable** (complications prevent an effective dose). Look for contributors: worsening liver disease, portal/hepatic-vein thrombosis, GI bleeding, infection/SBP, malnutrition, HCC, cardiac or renal disease, hepatotoxic or nephrotoxic substances.

E27.07 · Book p. 342 · Hepatology p. 74

Large-volume paracentesis, TIPS and shunting

HEPATOLOGY

Large-volume paracentesis: 4–6 L daily until the abdomen is emptied, repeated every 2–4 weeks; continue diuretics and sodium restriction afterwards. (Albumin replacement, 8 g per litre removed, is on book p. 342 and in the source.) TIPS lowers diuretic needs, renin and aldosterone but may precipitate encephalopathy or liver failure. Peritoneovenous shunting is another option.

E27.08 · Book p. 342 · Hepatology p. 74

Spontaneous bacterial peritonitis BOOK PP. 343–344

Who gets SBP, and which organisms

HEPATOLOGY

Infection of ascitic fluid may be spontaneous or follow a previous paracentesis. The **spontaneous** type develops in about **8% of cirrhotics with ascites** (the book's 10% is for all cirrhotic patients, a figure the source also gives), especially when the cirrhosis is severely decompensated. 90% are monomicrobial. More than one organism suggests paracentesis-related infection; anaerobes are rare. Variceal bleeding and previous SBP carry particular risk.

E27.09 · Book p. 343 · Hepatology pp. 75–76

From translocation to SBP

HEPATOLOGY

Gut bacteria translocate (overgrowth, dysmotility) to mesenteric nodes or bypass liver RES via portosystemic shunts → bacteraemia → **bacterascites**, which resolves if ascitic opsonic activity is good and becomes SBP if it is poor.

E27.10 · Book p. 343 · Hepatology p. 76 (figure)



SBP can be silent

HEPATOLOGY

Classic features (fever, pain, leucocytosis) **may be absent**; the book (p. 343) already says to suspect SBP in any cirrhotic with ascites who deteriorates.

E27.11 · Book p. 343 · Hepatology p. 76

Treatment length and prophylaxis after bleeding

HEPATOLOGY

Treat for ≥ 5 days; amoxicillin–clavulanate is as effective as third-generation cephalosporins. After upper GI haemorrhage (or sclerotherapy) give norfloxacin 400 mg 12-hourly for **at least 7 days**.

E27.12 · Book pp. 343–344 · Hepatology pp. 76–77

CHAPTER 28 AFTER MID-MODULE

↑ CONTENTS

Renal Impairment in Cirrhosis

Adds to book pp. 345–348 · book file 2 · book subjects: Internal Medicine

Hepatorenal syndrome BOOK PP. 346–347

The kidneys are normal: the defect is blood flow

HEPATOLOGY

The book (p. 346) gives the normal renal histology, renal vasoconstriction and splanchnic vasodilatation. The source adds that the kidneys are **capable of normal function** (HRS is progressive, and the defect is renal blood flow), that **cardiac dysfunction** is a third component, and that most patients have tense ascites.

E28.01 · Book p. 346 · Hepatology p. 78 (text + figure)

Classic criteria and what to exclude

HEPATOLOGY

Classic criteria used serum creatinine ≥ 1.5 mg/dL ($133 \mu\text{mol/L}$) persisting 48 h after diuretic withdrawal and albumin expansion. Exclude toxins (paracetamol, NSAIDs, carbon tetrachloride), infection, ATN (hypotension, aminoglycosides, contrast) and obstructive uropathy.

E28.02 · Book pp. 346–347 · Hepatology p. 79

Preventing HRS

HEPATOLOGY

Avoid volume depletion (diuretics, lactulose, GI bleeding, LVP without albumin); use ACEI/ARB/NSAIDs/antibiotics judiciously; treat infection early and give SBP prophylaxis; pentoxifylline in severe alcoholic hepatitis; prevent variceal bleeding.

E28.03 · Book p. 347 · Hepatology p. 80 (table)

Albumin loading and alternatives to terlipressin

HEPATOLOGY

- ◆ Albumin loading 1 g/kg/day (max 100 g) to a CVP of 10–15 cmH₂O.
- ◆ Alternatives to terlipressin: midodrine 2.5–5 mg tds (to 15 mg tds) + octreotide 100 μg SC tds (to 200 μg), or noradrenaline 0.1–0.7 $\mu\text{g/kg/min}$, titrated to MAP.
- ◆ Vasopressors generally for up to 2 weeks; evaluate for transplantation.

The terlipressin escalation schedule and maintenance albumin dose from the same table differ from the book and are kept for review (R36).

E28.04 · Book p. 347 · Hepatology p. 80 (table)



FOR REVIEW Source disagreements for this chapter are kept out of the teaching points above: **R36 Terlipressin escalation and albumin dose in HRS** (p. 45)

CHAPTER 29 AFTER MID-MODULE

[↑ CONTENTS](#)

Jaundice

Adds to book pp. 349–354 · book file 2 · book subjects: Internal Medicine

Approach to jaundice

 BOOK PP. 351–354

Isolated indirect hyperbilirubinaemia: the algorithm

HEPATOLOGY

First exclude recent trauma, haematoma or transfusion; then LDH, haptoglobin and reticulocyte count: abnormal → haemolysis/dyserythropoiesis; normal → Gilbert or Crigler–Najjar I/II.

E29.01 · Book p. 354 · Hepatology p. 10 (algorithm)

Halothane

HEPATOLOGY

Halothane (like paracetamol) causes toxic hepatitis with mixed (hepatic) hyperbilirubinaemia.

E29.02 · Book p. 351 · Hepatology p. 9

Biliary obstruction vs liver disease

HEPATOLOGY

	Favours biliary obstruction	Favours liver disease
History	Abdominal pain, fever/rigors, prior biliary surgery, older age	Viral prodrome, known viral exposure, blood products or IV drug use, hepatotoxin, family history
Examination	Fever, tenderness, palpable mass, surgical scar	Spider telangiectasias, stigmata of portal hypertension, asterixis
Laboratory	ALP ↑ out of proportion; PT normal or corrects with vitamin K; leucocytosis; amylase/lipase ↑	Aminotransferases ↑ out of proportion; PT does not correct with vitamin K; thrombocytopenia; positive serology

E29.03 · Book pp. 353–354 · Hepatology p. 87 (table)

Intrahepatic cholestasis by level

HEPATOLOGY

- ◆ **Hepatocyte:** viral, drug and alcoholic hepatitis; post-necrotic cirrhosis; endotoxin.
- ◆ **Canaliculus:** benign recurrent cholestasis; hormones, pregnancy, drugs.
- ◆ **Ductules:** PBC, allograft rejection, **graft-versus-host**, biliary atresia, idiopathic adult ductopenia.
- ◆ **Larger ducts:** primary/secondary sclerosing cholangitis.

E29.04 · Book p. 352 · Hepatology p. 85 (figure)



Hepatic Encephalopathy

Adds to book pp. 355–360 · book file 2 · book subjects: Internal Medicine

IN THIS CHAPTER Pathogenesis · Treatment framework

Pathogenesis BOOK PP. 358, 360

Ammonia

HEPATOLOGY

About half of gut ammonia is made by bacteria, the rest from dietary protein and glutamine; it is cleared as urea and glutamine. Blood ammonia is raised in **90%** of HE; ammonia damages the blood–brain barrier and astrocytes and shifts neural activity toward inhibition. Reduced muscle metabolism also raises it.

E30.01 · Book p. 358 · Hepatology p. 90 (text + figure)

Amino-acid imbalance and false neurotransmitters

HEPATOLOGY

Aromatic amino acids rise (failed hepatic deamination) while branched-chain amino acids fall (muscle/renal uptake driven by **hyperinsulinaemia**); more aromatic amino acids enter the brain, and phenylalanine inhibits dopa synthesis → false neurotransmitters.

E30.02 · Book pp. 358, 360 · Hepatology p. 90

GABA

HEPATOLOGY

GABA is made by gut bacteria and normally cleared by the liver; in liver failure or shunting it reaches the systemic circulation.

E30.03 · Book p. 358 · Hepatology p. 90

Treatment framework BOOK P. 359

Three strategies

HEPATOLOGY

Treat the **precipitating cause**; reduce production and absorption of gut-derived ammonia and toxins; modify neurotransmitter balance.

E30.04 · Book p. 359 · Hepatology p. 91

Acute Hepatic Failure

Adds to book pp. 361–366 · book file 2 · book subjects: Internal Medicine · Emergency / Acute Care

Cerebral oedema BOOK P. 363

Cerebral blood flow in acute liver failure

HEPATOLOGY

Cerebral blood flow is generally increased: persistent vasodilatation and loss of autoregulation cause hyperaemia, and the rise in cerebral blood volume can worsen brain oedema.

E31.01 · Book p. 363 · Hepatology p. 94



CHAPTER 32	Biliary System Anatomy NO ADDITIONS Book pp. 367–372 · book file 2 · Surgery Checked against GIT Others; the Hepatology source does not cover the biliary system.
CHAPTER 33	Gallbladder Diseases NO ADDITIONS Book pp. 373–375 · book file 2 · Surgery Checked against GIT Others; the Hepatology source does not cover the gallbladder.
CHAPTER 34	Gallstones and Cholecystitis NO ADDITIONS Book pp. 376–388 · book file 2 · Surgery Checked against GIT Others; the Hepatology source does not cover gallstones.
CHAPTER 35	Obstructive Jaundice NO ADDITIONS Book pp. 389–398 · book file 2 · Surgery · Radiology / Investigation The Hepatology source's cholestasis and jaundice material is filed under Chapters 18 and 29; biliary <i>Ascaris</i> (GIT Others) is already on book p. 390.
CHAPTER 36	Acute Pancreatitis NO ADDITIONS Book pp. 399–406 · book file 2 · Surgery · Emergency / Acute Care Checked against GIT Others; neither source covers acute pancreatitis.
CHAPTER 37	Cancer Pancreas NO ADDITIONS Book pp. 407–412 · book file 2 · Surgery · Pathology Checked against GIT Others; neither source covers pancreatic cancer.
CHAPTER 38	Acute Abdomen NO ADDITIONS Book pp. 413–417 · book file 2 · Emergency / Acute Care · Surgery Checked against GIT Others; neither source covers the acute abdomen.
CHAPTER 39	Abdominal Trauma NO ADDITIONS Book pp. 418–424 · book file 2 · Surgery · Emergency / Acute Care Checked against GIT Others; neither source covers abdominal trauma.
CHAPTER 40	Splenic Trauma NO ADDITIONS Book pp. 425–433 · book file 2 · Surgery · Emergency / Acute Care Checked against GIT Others; neither source covers splenic trauma.



SEPARATE FROM THE TEACHING POINTS

Points for Youssef to Review

These are disagreements found while comparing the three source handouts with the main book. Each point shows both (or all) positions with their page references. **None of them has been resolved here**, and none of the disputed values is used as a teaching point in the chapters.

Duplicate flags raised by more than one handout are merged. The ID in brackets under each point traces it to the original handout and ledger row.

CHAPTER 03 The Stomach and Duodenum · p. 6

R01 Autoimmune gastritis: is it "pangastritis"?

INTERNAL MEDICINE

Book pp. 38, 42

Autoimmune gastritis affects the fundus and body and **spares the antrum**; "pangastritis" is used only for H. pylori gastritis spreading from antrum to body.

GIT Internal Med p. 26

Autoimmune gastritis affects the body and fundus and is called **"pangastritis"**.

Raised by: GIT Internal Med companion C3-13

R02 Misoprostol: a PGE₁ or a PGE₂ analogue?

PHARMACOLOGY

Book p. 37

"a synthetic prostaglandin **E₂** analogue".

Book p. 48 · GIT Others p. 4

"Synthetic **PGE₁** analog".

The book's two pages disagree with each other; GIT Others agrees with p. 48.

Raised by: GIT Others capsule C3-D1 (ledger PH-15)

R03 H. pylori triple therapy: duration and first-line status

INTERNAL MEDICINE

Book p. 41

PPI + amoxicillin + clarithromycin for **14 days**, but **no longer recommended as empirical first-line** unless local clarithromycin resistance is known to be low.

Book p. 48

The same regimen for **10–14 days**.

GIT Internal Med p. 31

The same regimen for **14 days**, "one of the most commonly used" regimens.

The book is inconsistent on duration (14 vs 10–14 days), and the sources differ on whether triple therapy is still first-line.

Raised by: GIT Internal Med companion C3-17

R04 Levofloxacin dose in the rescue regimen

PHARMACOLOGY

Book p. 41

"Levofloxacin **500 mg once daily** + PPI + Amoxicillin".

Book p. 48 · GIT Others p. 5

"levofloxacin **250 or 500 mg twice a day** for 10 days" (with PPI + amoxicillin 1 g twice a day).

The book's two pages disagree with each other; GIT Others agrees with p. 48.

Raised by: GIT Others capsule C3-D2 (ledger PH-16)

R05 Transfusion threshold in acute upper GI bleeding

INTERNAL MEDICINE

GIT Internal Med p. 39

Transfuse for shock **or Hb < 10** with recent or active bleeding. The source prints the unit as "mg/dl".

Book pp. 50–51

"TV fluids and blood transfusion **when indicated**"; "Hemoglobin levels are generally a **poor indicator** of the need to transfuse". No numeric threshold is given.

For variceal bleeding the book (Ch 23, p. 315) uses a restrictive target of Hb 7–8 g/dL (see R32). The shock criteria from the same source page are taught in Chapter 3; the Hb threshold is not.

Raised by: Integration review of GIT Internal Med companion ledger row GB-08



R06 Commonest site of gastric cancer: how common?

PATHOLOGY

Book p. 62

Antrum/pyloric area **60%**, body 15%, fundus 25%.

GIT Others p. 15

Pyloric region **50%**, then **lesser curvature 25%**; it can occur anywhere in the stomach.

Both agree that the pyloric (antral) region is the commonest site; the percentages and the second site differ.

Raised by: GIT Others capsule C3-D3 (ledger TU-13)

CHAPTER 05 Diarrhea and Malabsorption · p. 12

R07 How long is "acute" diarrhoea?

INTERNAL MEDICINE

Book p. 77

Acute diarrhoea lasts **less than 2 to 3 weeks or, rarely, 6 to 8 weeks**.

GIT Internal Med p. 51

Acute diarrhoea lasts **less than 4 weeks**.

Both agree that diarrhoea lasting more than 4 weeks is evaluated as chronic.

Raised by: GIT Internal Med companion C5-08

R08 Stool fat and lactose breath test protocols

INTERNAL MEDICINE

Book p. 93 · GIT Internal Med p. 68 (text)

70-100 g fat diet with stool collected over **2-3 days**; lactose dose **25 g**.

GIT Internal Med pp. 69-70 (tables)

100 g fat diet with stool collected over **3 days**; lactose dose **1 g/kg** body weight.

The source's text and tables disagree with each other; the book follows the text.

Raised by: GIT Internal Med companion C5-38

CHAPTER 06 Constipation · p. 17

R09 IBS diagnostic criteria: which Rome edition?

INTERNAL MEDICINE

Book p. 102 (Rome V)

Pain on average **at least 3 days per month** over the last 3 months.

GIT Internal Med p. 75 (Rome IV)

Recurrent abdominal pain **at least 1 day per week** in the last 3 months.

The 2-of-3 associated features (related to defecation, change in stool frequency, change in stool form) are the same in both.

Raised by: GIT Internal Med companion C6-16

R10 Low-FODMAP diet: which foods to limit?

INTERNAL MEDICINE

Book p. 104

Foods high in fructans and oligosaccharides: **onions, garlic, wheat, rye and legumes**.

GIT Internal Med p. 75

Potatoes, brown rice, oats and almonds as examples of foods the patient should limit.

Raised by: GIT Internal Med companion C6-17

R11 IBS-C: the second drug option

INTERNAL MEDICINE

Book p. 105

Prucalopride (selective 5-HT₄ agonist, prokinetic). Linaclotide appears only in the laxative table on p. 99, for refractory constipation.

GIT Internal Med p. 75

Linaclotide, a guanylate cyclase activator that softens stool and stimulates bowel movement.

Both give lubiprostone (chloride channel activator) as the first option.

Raised by: GIT Internal Med companion C6-20



CHAPTER 07 Foodborn Infection · p. 18

R12 Widal test: what a high H titre means

MICROBIOLOGY

Book p. 109

A high titre to **one** H antigen ($\geq 1:160$) suggests recent infection with that type; high O titre **plus more than one** H antigen indicates recent immunisation.

GIT Others p. 24

"**More than one** H antigen ($\geq 1:160$) suggest recent infection with this type of Salmonella **and** recent immunization."

The GIT Others list here is garbled (its items 1–2 are cut off), so it is not a reliable correction.

Raised by: GIT Others capsule C7-D1 (ledger FI-13)

CHAPTER 08 Inflammatory Bowel Disease · p. 19

R13 Ulcerative colitis: two rows of GIT Internal Med's p. 76 table vs the book's Fig 8.1

INTERNAL MEDICINE

Book p. 117 (Fig 8.1)

Granulomas **should not be present** in UC (microgranulomas may accompany ruptured crypt abscesses); GIT Internal Med's own p. 77 table agrees. Rectum: "**typically involved**" in UC.

GIT Internal Med p. 76 table

Granulomas in UC: "**occasional**". Rectum: "**always**" involved in UC (only commonly in Crohn's disease).

Raised by: GIT Internal Med companion C8-03 (granulomas); rectum row found in integration review of C8-02

R14 Systemic steroid doses in IBD

INTERNAL MEDICINE

Book pp. 119–120 · GIT Internal Med p. 80 (text)

Methylprednisolone **10–30 mg IV q6–8h**; prednisone **40–60 mg/day**.

GIT Internal Med p. 80 (drug table)

Methylprednisolone **40–60 mg IV daily**; prednisone **0.25–0.75 mg/kg PO daily**.

Raised by: GIT Internal Med companion C8-17

CHAPTER 15 Colorectal Cancer · p. 22

R15 Colorectal cancer: age at presentation

PATHOLOGY

GIT Others p. 19

"Age of **50–70** years."

GIT Others p. 20 · Book p. 74 (Ch 4)

"between **60 and 79** years of age".

Book p. 196 (Ch 15)

"old age, however no age is immune".

GIT Others contradicts itself.

Raised by: GIT Others capsule C15-D1 (ledger TU-31)

CHAPTER 18 Liver Disease Assessment · p. 23

R16 PBC in the immunoglobulin list: raised AMA or raised IgM?

HEPATOLOGY

Book p. 239

Under "Serum immunoglobulins": "In primary biliary cirrhosis, **raised IgM**."

Hepatology p. 11

Same list: "In primary biliary cirrhosis, **raised AMA**."

Both texts give AMA 95% under autoantibodies.

Raised by: Hepatology extras ledger A-34

R17 What happens to the reabsorbed 10–20% of urobilinogen?

HEPATOLOGY

Book p. 245

The reabsorbed 10–20% **escape the enterohepatic circulation** and pass to the kidney to be excreted in urine.

Hepatology p. 82

The reabsorbed 10–20% enter the portal blood and are **re-excreted by the liver**; only a small fraction escapes hepatic uptake into urine (figure: ~80% faeces, ~18% enterohepatic, ~2% urine).

Raised by: Hepatology extras ledger F-04



R18 Gilbert's syndrome: decreased uptake or decreased conjugation?

HEPATOLOGY

Book p. 245 (Ch 18)

Gilbert's is listed under "II – Decrease uptake" ("defect of hepatic uptake and/or decrease of conjugation").

Book pp. 350–351 (Ch 29) · Hepatology pp. 83–84

Gilbert's is a **decreased conjugation** disorder (deficient hepatic bilirubin glucuronidation); drugs such as rifampin are the example of decreased uptake.

The book's two chapters classify it differently.

Raised by: Hepatology extras ledger F-05

CHAPTER 19 Acute Hepatitis · p. 27

R19 Rest in acute viral hepatitis

HEPATOLOGY

Book p. 259 (also p. 253 for HAV)

"Normal activity as tolerated"; for HAV, "rest and dietary measures are unhelpful".

Hepatology p. 31

"Bed rest is **mandatory** until the patient is free of jaundice and symptoms and the bilirubin is less than 1.5 mg/dl."

The Hepatology source itself also says rest is unhelpful for HAV (p. 24).

Raised by: Hepatology extras ledger B1-35

CHAPTER 20 Chronic Hepatitis B · p. 28

R20 HBV prevalence in Egypt

HEPATOLOGY

Book p. 265

"currently estimated to be **below 2%** in the general population"; genotype D predominant.

Hepatology p. 35

Overall prevalence in Egypt **5%**, mainly genotype D.

Raised by: Hepatology extras ledger B2-03

R21 Risk of chronic HBV by age at infection

HEPATOLOGY

Book p. 265 (and p. 247)

Neonates ~90%, children under 5 years **20–30%**, adults **<5%** (p. 247: chronic hepatitis in 5–10%).

Hepatology p. 35

Neonates 90%, children under 5 years **20–50%**, adults **<10%**.

Raised by: Hepatology extras ledger B2-04

R22 Inactive carrier: HBV DNA level

HEPATOLOGY

Book pp. 266–267

HBV DNA usually **<2,000 IU/mL**.

Hepatology p. 35

HBV DNA **below 400 IU/L**.

The source's unit (IU/L) is unusual.

Raised by: Hepatology extras ledger B2-10

R23 Spontaneous HBsAg clearance per year

HEPATOLOGY

Book p. 267

Approximately **1%** per year.

Hepatology p. 35

1–2% per year.

Raised by: Hepatology extras ledger B2-12

R24 Egypt's HCV screening campaign: how many screened?

HEPATOLOGY

Book p. 273

Approximately **60 million** people have been screened; no prevalence figure.

Hepatology p. 39

More than **49 million** screened (Oct 2018–May 2019), HCV prevalence **4.4%**.

The two may refer to different time frames.

Raised by: Hepatology extras ledger B2-30



CHAPTER 21 Autoimmune Hepatitis · p. 30

R25 How common is type 1 autoimmune hepatitis?

HEPATOLOGY

Book p. 275

Type 1 accounts for approximately **80–90%** of cases.

Hepatology p. 22 (table)

Type 1 **95%**, type 2 **5%**.

Raised by: Hepatology extras ledger B1-24

R26 Metformin in fatty liver disease

HEPATOLOGY

Book p. 279

Metformin should be used to treat diabetes but is **not recommended specifically for treating MASH**; pioglitazone may be considered.

Hepatology p. 44

Insulin-sensitising agents "such as **metformin** or pioglitazone" are listed for NAFLD.

Raised by: Hepatology extras ledger C-02

CHAPTER 22 Liver Cirrhosis · p. 30

R27 Feto hepaticus: what it smells like

HEPATOLOGY

Book p. 283

"a **sweet, musty** odor caused by volatile sulfur-containing compounds".

Hepatology p. 46

"a **fecal** smell due to lack of detoxification of mercaptan absorbed from colon".

Raised by: Hepatology extras ledger C-08

R28 Dietary protein in hepatic encephalopathy

HEPATOLOGY

Book p. 286 (Ch 22)

Protein restriction is no longer recommended: **1.2–1.5 g/kg/day** even in HE; temporary reduction only in severe overt HE.

Hepatology p. 48

In encephalopathy, proteins should be **restricted to 30–40 g/day**.

Book p. 359 (Ch 30)

"In the acute attack dietary protein is reduced to **20 g/day**", with the book's own note that this is an unresolved source claim.

Raised by: Hepatology extras ledger C-13; third position (book p. 359) found by the integration audit

R29 PBC: how many diagnostic criteria are needed?

HEPATOLOGY

Book p. 288

Two of three: cholestatic ALP, AMA, compatible biopsy.

Hepatology p. 51

All three are "**mandatory**": cholestasis, M2 antibodies, destructive cholangitis on biopsy.

Raised by: Hepatology extras ledger C-15

R30 UDCA in PBC: does it change the disease?

HEPATOLOGY

Book p. 288

UDCA is first-line; it improves biochemistry, **slows fibrosis progression and improves survival**.

Hepatology p. 51

UDCA improves bilirubin and liver biochemistry but has **no effect on the disease process**.

Raised by: Hepatology extras ledger C-16

R31 Alcohol threshold for alcoholic cirrhosis: grams or millilitres?

HEPATOLOGY

Book p. 281 (also p. 298: >10 years, no amount) · Hepatology p. 44

Regular intake **>30 ml/day** for more than 10 years.

Hepatology p. 55

At least **30 g** of alcohol daily for more than 10 years.

30 ml of ethanol is not 30 g.

Raised by: Hepatology extras ledger C-26



CHAPTER 23 Portal Hypertension · p. 32

R32 Transfusion target in acute variceal bleeding

HEPATOLOGY

INTERNAL MEDICINE

Book p. 315

Transfusion should be **restrictive**, with a target haemoglobin generally around **7–8 g/dL**.

Hepatology p. 61 · GIT Internal Med p. 43

Over-transfusion may cause rebleeding; keep the **haematocrit around 30**.

Both warn against over-transfusion; the numeric targets differ.

Raised by: Hepatology extras ledger D-24; the same text is in GIT Internal Med p. 43 (companion row GB-18, out of its scope)

R33 Nasogastric (Ryle's) tube in GI bleeding: routine or selective?

HEPATOLOGY

INTERNAL MEDICINE

Book p. 315 (variceal bleeding)

A nasogastric tube may be considered **selectively** but is **not routinely required** in every patient.

Hepatology p. 61 (variceal bleeding)

Ryle's tube insertion for aspiration and lavage, listed as part of **routine** resuscitation.

GIT Internal Med pp. 40, 46 (non-variceal and lower GI bleeding)

A nasogastric tube for gastric wash and monitoring of ongoing bleeding; nasogastric lavage in the unstable patient with lower GI bleeding.

The book's Chapter 3 bleeding pages (pp. 50–53) do not mention a nasogastric tube. The GIT Internal Med uses are kept in Chapter 3 with this cross-reference.

Raised by: Hepatology extras ledger D-25; GIT Internal Med companion GB-09 and GB-25 (context)

R34 Screening endoscopy for every cirrhotic?

HEPATOLOGY

INTERNAL MEDICINE

Book pp. 285, 313

Assessment by non-invasive criteria and/or upper GI endoscopy; guided by **Baveno VII**, some patients can **safely avoid endoscopy**.

Hepatology p. 60 · GIT Internal Med p. 43

"Every patient diagnosed with cirrhosis should have an upper GI endoscopy".

The Hepatology source itself (p. 60) also gives the elastography criteria that spare endoscopy (taught in Chapter 22).

Raised by: Hepatology extras ledger D-26; the same sentence is in GIT Internal Med p. 43

CHAPTER 25 Pyogenic Liver Abscess · p. 33

R35 Blood count in amoebic liver abscess

PARASITOLOGY

Book p. 331 (Ch 25)

"CBC: Leukocytosis and **eosinophilia**".

Book p. 138 (Ch 9) · GIT Others p. 47

CBC shows **polymorphonuclear leucocytosis**, which helps separate amoebic liver abscess from viral hepatitis. Neither mentions eosinophilia.

Both agree on leucocytosis.

Raised by: GIT Others capsule C25-D1 (ledger PA-29)

CHAPTER 28 Renal Impairment in Cirrhosis · p. 36

R36 Terlipressin escalation and albumin dose in HRS

HEPATOLOGY

Book p. 347 · Hepatology text

Terlipressin 1 mg q4–6h or 2 mg/day infusion; escalate if creatinine falls <25% **after 2 days**, to a maximum of 12 mg/day; albumin 20% at **20–40 g/day**.

Hepatology p. 80 (image table)

Start 1 mg IV every 4 h; increase up to 2 mg IV every 4 h if creatinine has not improved by 25% **at day 3**; maintenance albumin **20–60 g/day**.

Both versions appear in the Hepatology source (text vs image table); the book carries only the text version.

Raised by: Hepatology extras ledger E-37

ALSO NOTED Wording and consistency notes (not used as teaching points)

#	Where	What is printed
N1	Book p. 250 (Ch 18) and p. 297 (Ch 22)	Wilson disease hepatic copper: "≥250 mcg/g dry weight" on p. 250, ">250 µg/g dry weight" on p. 297.



#	Where	What is printed
N2	Book p. 271 (Ch 20)	HCV chronicity: "55–85% develop chronic infection" and "chronic hepatitis in 80% of cases" on the same page.
N3	Book p. 359 (Ch 30) · Hepatology p. 91	Rifaximin "alters the intestinal flora <i>with any systemic side effects</i> " (as printed in both).
N4	Hepatology p. 62	Terlipressin: "It should be given to patients with ischemic heart disease" (as printed). This sentence is not used in this handout.
N5	Hepatology p. 49	Child–Pugh: class B "7–9 points", class C "> 10 points", which leaves a score of 10 unclassified.

HOW THIS HANDOUT WAS MADE

Sources, method and traceability

The three handouts merged here

Handout	Compared source	Scope	Subject labels here
Before Mid-Module Internal Medicine Extras	GIT Internal Med (82 pp.)	Chapters 1–15	Internal Medicine
Other Subjects Capsule	GIT Others (47 pp.)	All 40 chapters	Pharmacology, Pathology, Microbiology, Parasitology
After Mid-Module Hepatology Extras	Hepatology (96 pp.)	Chapters 16–40	Hepatology

This handout was rebuilt from the three handouts' editable sources and comparison ledgers, not by joining their PDFs. Each retained point keeps its meaning and its page references. The label on a point names the source it came from (GIT Internal Med, GIT Others by subject, or the Hepatology source).

Page references

- **Book p.** is the printed page of the main book, *YMnotes Digestive System, Version 3*. File 1 (4_5875397832227168769.PDF) holds printed pp. 1–197, where printed page = PDF viewer page – 3. File 2 (4_5875397832227168769(1).PDF) holds printed pp. 197–433, where printed page = PDF viewer page + 196; its built-in viewer labels restart at 1 and should not be used.
- Printed p. 197 appears in both files. Chapter 15 (Colorectal Cancer, pp. 195–209) spans the join and is treated as one chapter.
- Source pages (GIT Internal Med, GIT Others, Hepatology) are the printed page numbers of those files, which equal their PDF pages.
- The main book does not mark a mid-module boundary. Part 1 (Chapters 1–15) and Part 2 (Chapters 16–40) follow the scope of the Before- and After-Mid-Module handouts.

What the integration did

- **149 teaching additions** in 22 chapters: 52 Internal Medicine, 1 Pharmacology, 12 Pathology, 6 Microbiology and 78 Hepatology. Together they carry 254 original ledger rows. 18 chapters have no additions and appear as short entries.
- Points that taught the same topic from the same book page were combined into one entry, with every distinct fact and every page reference kept. The three handouts contained no true duplicates of each other: where two touch the same topic (for example, the *H. pylori* → cancer sequence in Chapters 3 and 4), each keeps its own facts and they refer to each other.



- One misfiled point (Hepatitis G, book p. 249) was moved from Chapter 19 to Chapter 18, where that page belongs.
- Sentences that an independent check found already on the cited book pages were trimmed. Two values that the source handouts printed as settled but that the book contradicts were moved out of the teaching points into review: the Hb < 10 transfusion threshold (R05) and "rectum always involved in UC" (R13). Where a point needs the book's own statement beside it to be read safely (nasogastric tubes, TIPS timing, albumin with large-volume paracentesis, a single-source terlipressin dose), that statement is printed with it.
- **36 points for review:** the 15 disagreements printed in the Internal Medicine and Other Subjects handouts, the 20 recorded only in the Hepatology ledger, and one found during integration (the transfusion threshold in upper GI bleeding). Two Hepatology points were confirmed on GIT Internal Med p. 43 and now cite both sources. None of them is resolved here, and no disputed value is used as a teaching point.

Traceability

Every entry and review point in this PDF carries an ID (E03.12, R05 ...). The merged ledger in the *ledger/* folder of the integration branch lists each one with the original handout, the original entry and ledger row IDs, the book and source pages, and what the integration did with it. It also carries every row the three handouts had already excluded, and the reason. The editable source and build instructions are in the same folder.

Limits

- This handout adds only what the three source handouts found. It does not re-compare the source PDFs with the book from scratch.
- The Internal Medicine comparison covered book pp. 195–197 of Chapter 15 only. GIT Internal Med has no colorectal-cancer section. Its variceal-bleeding pages (pp. 42–44) were outside that handout's scope; the Hepatology source carries the same text and was compared with Chapter 23.
- Additions reflect the teaching sources, not a clinical update. Doses and regimens are reproduced as printed and should be taught alongside the book.

رَبِّ زِدْنِي عِلْمًا

[سورة طه: 114]



محاضرات فيديو مراجعة شاملة

COMPLETE MEDICAL VIDEO LECTURES



ENT



Forensic
Medicine



Special Sense
Eye



Ophthalmology



Rheumatology



Orthopedic



Clinical
Toxicology



Cardiovascular
System



Cardiology



Respiratory
System



Chest



Gastrointestinal
Tract



General
Surgery



Special
Surgery



Urinary
System



Pediatric 1



Pediatrics 2



Surgery



Neurology



Endocrinology



Obstetrics



Nephrology



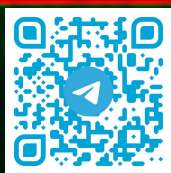
GIT



Vascular



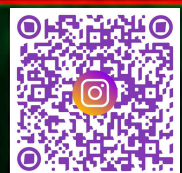
YouTube



Telegram



Facebook



Instagram



شرح مبسط
ومنظم



محتوى شامل
ومحدث



مناسب لجميع
مراحل الدراسة



راجع في أي وقت
ومن أي مكان